



MINI PROJECT FINAL REPORT

Predicting Rural Water Point Functionality Using Data Mining

Fundamentals of Data Mining (IT3051)

Group Mini Project

WaterPoint Intelligence: Rural Water Point Functionality Decision Support System

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Abstract

Rural water-point failure can interrupt access to safe water, while routine inspection resources are limited. This project investigates whether survey and contextual attributes can support the prioritisation of field assessment through three-class functionality prediction. The OpenWashData Rural Water Point Functionality dataset contains 1,793 observations across nine countries and 52 raw variables. The target distinguishes Functional, Partially functional, and Abandoned or not functional water points. Since the Functional class represents 74.3% of records, Macro-F1 and per-class outcomes were prioritised over accuracy alone.

A leakage-safe workflow removed contemporary status indicators and identifiers before a stratified 80/20 split into 1,434 development records and a locked 359-record holdout. Five model families were compared with repeated stratified cross-validation. The three strongest families, HistGradientBoosting, XGBoost and Random Forest, advanced to optimisation. The final Random Forest combines base tree preprocessing, rare-category frequency 10, mutual-information SelectKBest with 60 processed features, and no imbalance treatment. It was selected from development evidence before the holdout was evaluated. Its repeated-CV Macro-F1 was 0.6996 ± 0.0481 ; one-time holdout Macro-F1 was 0.6699, with accuracy 0.8663 and MCC 0.6466.

The model is exported as a verified serialised inference pipeline. A FastAPI service validates the 32 raw inputs and returns class probabilities; a Next.js dashboard presents predictions and exported analysis evidence. The principal limitation is poor recognition of Partially functional points: holdout recall was 0.1935 and F1 was 0.3000. Post-hoc geographic checks also show weaker transfer across countries. Therefore, the application is presented as decision support for inspection and maintenance planning rather than an automated determination of field condition.

Keywords: rural water points; multiclass classification; Random Forest; class imbalance; data leakage; decision support.

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CV	Cross-validation
EDA	Exploratory Data Analysis
F1	Harmonic mean of precision and recall
MCC	Matthews correlation coefficient
OOF	Out-of-fold
RF	Random Forest
SMOTENC	Synthetic Minority Over-sampling Technique for nominal and continuous features

1. Introduction

1.1 Background

A water point may appear serviceable in asset inventories even when its present condition has deteriorated. Limited field capacity makes prioritisation important. This project uses historical survey records to estimate functionality status from attributes potentially available before a visit.

A nonfunctional point can impose travel and repair costs on communities, while a partly functional point may still deliver water intermittently. The project therefore preserves all three statuses instead of collapsing them into a binary outcome. The dataset documents observations from nine countries, so a result must be interpreted with attention to location and survey variation.

1.2 Problem Statement

The three target states have operationally different meanings. A high overall accuracy can conceal missed partially functional sites because that class is small.

The operational question is whether an organisation can use attributes known before a field visit to identify points needing closer assessment. A model trained on present-day availability or recent breakdown questions would answer a different, easier question because those fields participate in the recorded outcome. This distinction shaped the leakage audit and the final input contract.

1.3 Aim and Objectives

The aim was to build, evaluate and expose a reproducible three-class prediction workflow. Objectives were to understand the dataset; prevent leakage; compare model families; select on development CV; evaluate once on holdout; and deliver an integrated decision-support interface.

Success was assessed through evidence at each stage: clean source profiling, reproducible preprocessing, comparison of more than the assignment's minimum four algorithms, development-only selection, a one-time holdout check, and a working prediction interface. These objectives correspond to the assignment workflow rather than to a single headline accuracy.

1.4 Scope

The system accepts 32 raw water-point attributes covering relevant infrastructure, geographic, operational, and management characteristics. It processes these inputs using the trained preprocessing pipeline and final classification model, returning a predicted functionality class together with probabilities for all three classes. The system is intended as a decision-support tool for identifying potential functionality issues and prioritising further inspection. It does not verify the actual physical condition of an individual water point, diagnose specific faults, or prescribe maintenance or repair actions.

1.5 Project Contributions

The completed work includes documented EDA, family-specific preprocessing, a locked model-selection protocol, production serialization checks, an API, and a browser dashboard.

1.6 Stakeholders and Decision Support

Potential users include water authorities, NGOs, maintenance teams and researchers. Predictions can help rank inspection candidates alongside local knowledge and field records.

The system can assist NGOs, public water authorities and maintenance teams with prioritisation. A high risk score can motivate inspection, but the result cannot establish physical condition without observation. Table 1.1 links the users identified in the proposal to concrete decisions and safeguards.

Table 1.1. Stakeholder requirements and responsible use

Stakeholder	Supported decision	Input or output	Safeguard
Water authority	Prioritise inspection	Status and class probabilities	Field verification
NGO / maintenance team	Plan rehabilitation visits	Site characteristics and risk ranking	Review local context
Researcher	Study comparative patterns	Exported EDA and model evidence	Avoid causal claims

1.7 Report Organisation

Chapters 2–9 cover the dataset, data analysis, modelling process, and supporting evidence; Chapters 10–11 present the implemented system and its testing; and Chapters 12–16 discuss the findings, practical implications, limitations, and directions for future improvement.

2. Background and Related Concepts

2.1 Rural Water-Point Functionality Monitoring

Functionality is a practical service indicator, but surveys are intermittent and heterogeneous across settings. A prediction can direct attention to uncertain or high-risk assets.

2.2 Water-Point Functionality as a Data-Mining Task

The recorded target is a nominal three-class label. Predictors mix geography, technology, governance, usage, maintenance and environment.

2.3 Multiclass Classification and Class Imbalance

Macro-F1 averages class F1 equally, making the rare Partial class visible. Accuracy remains a supporting measure because the majority class dominates it.

The majority Functional class contains 1,333 of 1,793 points. A constant majority-class classifier would already obtain 74.3% accuracy, demonstrating why accuracy alone is insufficient. Macro-F1 gives each class equal weight; balanced accuracy and class-level recall expose missed minority cases. MCC offers a complementary single-number measure that uses all cells of the confusion matrix.

2.4 Data Leakage

Variables describing current water availability or recent breakdowns may directly reflect the target outcome. Including them could cause data leakage and produce unrealistically high performance for pre-inspection prediction.

This risk is especially important because the target is defined using water availability, water quantity, and recent downtime. Dataset_proposal After examining their relationship with the target, leakage-prone variables were excluded before model training, preventing direct target proxies from artificially improving baseline results.

2.5 Cross-Validation and Holdout Evaluation

Repeated stratified folds estimate variation during development. A previously untouched holdout gives one final historical check after configuration lock.

Repeated stratified folds preserve class proportions while sampling different validation partitions. This reduces dependence on one lucky split, although folds are not independent observations. The locked historical holdout is a separate check: it was not used to choose hyperparameters, rare-category thresholds or selected features.

2.6 Classification Algorithms Considered

Logistic Regression supplies a linear reference; Random Forest and gradient boosting capture nonlinear mixed-feature relationships. XGBoost and CatBoost add alternative boosted-tree approaches.

Figure 2.1 positions the five-family baseline, three-family finalist stage, final lock and later robustness work within the project lifecycle. The algorithms differ in their sensitivity to scaling, nonlinear interactions and categorical representation; Table 2.1 states the role each served.

Table 2.1. Candidate families and their methodological roles

Family	Role	Representation
Logistic Regression	Linear, scale-sensitive reference	Scaled / engineered winner
Random Forest	Nonlinear bagged trees	Tree base
HistGradientBoosting	Histogram-based boosted trees	Tree base
XGBoost	Regularized boosted trees	Tree base
CatBoost	Boosting with native categorical route	Native categorical base

Random Forest follows the bagged-tree formulation [6]; XGBoost and CatBoost were assessed as distinct boosted-tree families [7], [8]. The repeated-stratified validation and mutual-information selector follow the documented estimators [9], [10].

2.7 Project Data-Mining Workflow

The sequence was data understanding, EDA, fixed exclusions, split, fold-fitted preprocessing, baseline comparison, development-only optimization, final lock, holdout evaluation and post-hoc robustness.

Project lifecycle

The post-hoc stage followed the final model lock.

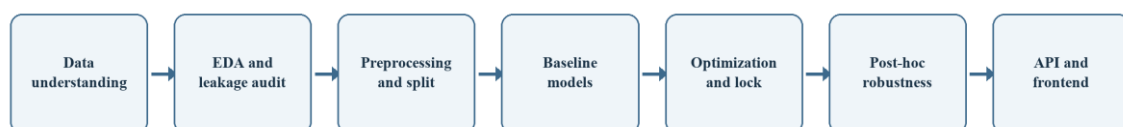


Figure 2.1. Project lifecycle from data understanding through post-hoc review and integrated delivery

3. Dataset Description and Validation

3.1 Source and Licensing

The project repository identifies the Rural Water Point Functionality dataset (rwpfunctionality) from OpenWashData [1]. The dataset is distributed under CC-BY according to project records. The raw workbook was retained as a read-only source during analysis.

Table 3.1. Dataset profile

Property	Recorded value
Rows	1793
Raw variables	52
Countries	9
Target	functional3
Classes	3
Columns containing missing values	40
Fully duplicate rows	0

3.2 Dataset Profile

The sample spans nine countries but is modest relative to the diversity of water-point technologies and local conditions.

Figure 3.1 shows the accounting from 52 raw variables to the 32-field production contract. The target and excluded fields are not exposed as inputs. The retained groups preserve information about site location, infrastructure, usage, governance and environment.

3.3 Target Variable and Class Meaning

Functional denotes serviceable points; Partially functional denotes degraded service; Abandoned or not functional denotes unavailable service. Labels are taken as recorded in the dataset.

The proposal defines the labels through availability, sufficiency and recent downtime. Those criteria make the three classes meaningful for operations, but they also explain why the corresponding survey questions cannot be used as prospective predictors. Table 3.2 makes this relationship explicit.

Table 3.2. Target classes, recorded meaning and modelling implication

Class	Dataset interpretation	Modelling implication
Functional	Water available and sufficient, without recent downtime	Majority class; high accuracy can be misleading
Partially functional	Water available but insufficient or recently interrupted	Rare, operationally important middle state
Abandoned / not functional	No water under recorded criteria	Clear failure class but still has diverse contexts

3.4 Predictor Categories

The final input schema contains 32 raw attributes covering location, water-point technology, rehabilitation history, management, committee capacity, usage, and environmental characteristics..

Table 3.3 maps representative raw columns to the five frontend groups. This is an interface organisation, not a change to the model schema. The exact 32 names remain the authority for the API and are listed in Appendix B.

Table 3.3. Production input groups and representative fields

Group	Example raw predictors	Interface purpose
Location	country, admin1, latitude_wp, longitude_wp	Situate the asset
Infrastructure	wptype, pumptype, drillmethod, wp_age	Describe construction and history
Usage / demographics	qtypepeople_wp, qtyhh_wp, pop_1000	Describe demand context
Management / finance	whomanage_wp, wc_*_index_wp, wc_savings_wp	Describe governance
Environment	annual_rain, season	Describe external context

Dataset to production inputs

Location | infrastructure | usage | management and finance | environment

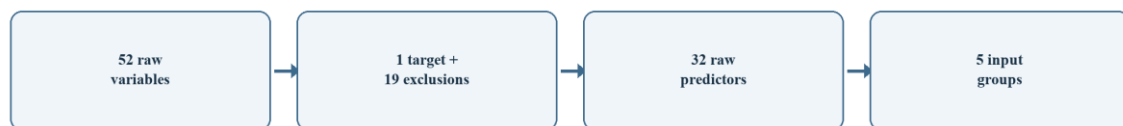


Figure 3.1. Accounting from 52 raw variables to five groups of 32 production predictors

3.5 Excluded Attributes and Rationale

Seven current-status fields were excluded for leakage. Ten identifiers or high-cardinality administrative fields were excluded. Two pending financial fields were not used. The final retained set contains 32 raw predictors.

3.6 Dataset Suitability and Instructor Validation

The dataset proposal records the OpenWashData source, 1,793 records, the three-class target, CC-BY licence, stakeholder setting and anticipated leakage risks. It establishes why the dataset suits the assignment's multiclass workflow. The assignment requires instructor validation before development; the proposal is available, but a separate approval record was not supplied for this report.

3.7 Initial Data-Quality Observations

Forty of 52 columns contain some missing values; zero complete duplicate rows were recorded. Missingness has different meanings across survey branches.

Table 3.4. Feature exclusions

Category	Excluded variables
Current-status leakage	wateravailable; whynowatertoday_wp; whynowatertoday_wp_other; downtime2weeks; minutesfill20l; brokendays_wp; qtymonthsnwater_wp

Identifiers / administration	id; instance_wp; photo_wp; commid_wp; cwid; timepoint_wp; dataorg_wp; subdate_wp; admin2; admin3
Pending / unused	grant_number_wp; balance_any_dollars_wp

4. Exploratory Data Analysis

4.1 EDA Approach

Notebook 02 profiled the unmodified raw dataset before learned preprocessing or model selection.

4.2 Target Distribution and Class Imbalance

Functional accounts for 1,333 records (74.3%), Partially functional 157 (8.8%), and Abandoned or not functional 303 (16.9%). Figure 4.1 makes the imbalance clear.

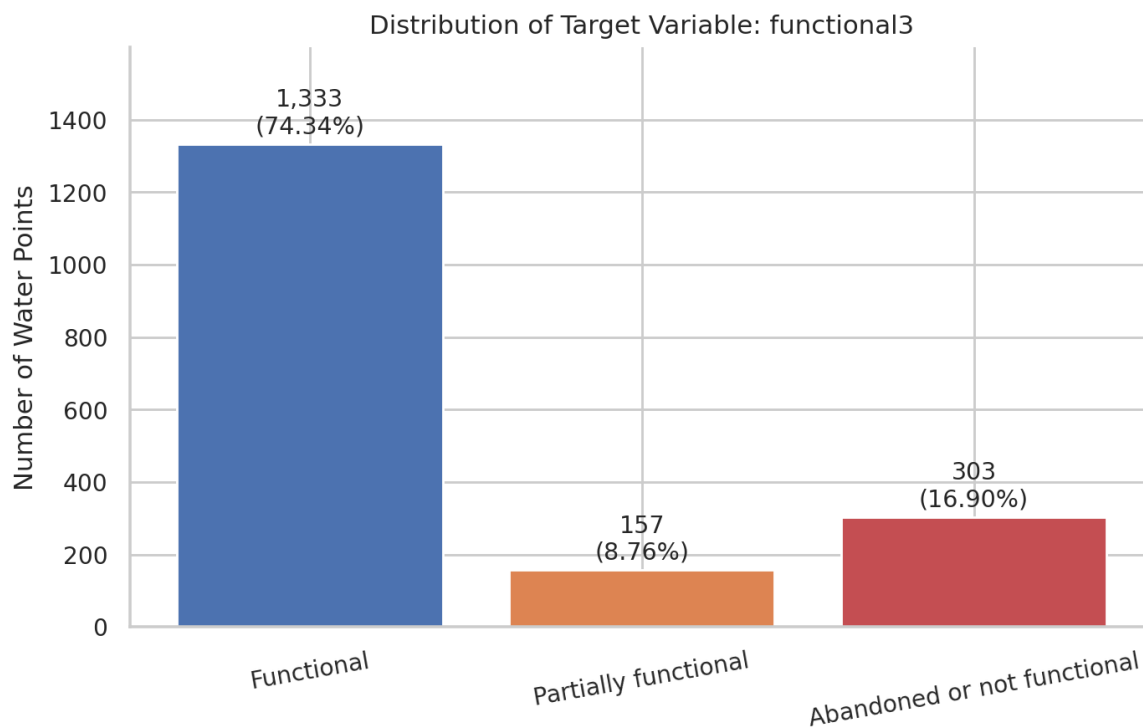


Figure 4.1. Target class distribution (Source: project export)

The raw label distribution is already an evaluation warning: 1,333 Functional observations account for 74.3% of records, while the Partial class contains 157 and the Abandoned class 303. Stratification preserves the approximate shares in development and holdout partitions, but it cannot create information that the rare class does not contain. The class chart establishes why subsequent plots and metrics retain all three labels.

4.3 Missing-Value Profile

Several fields are conditionally relevant; others are absent in survey batches. A single blanket fill value would erase this distinction.

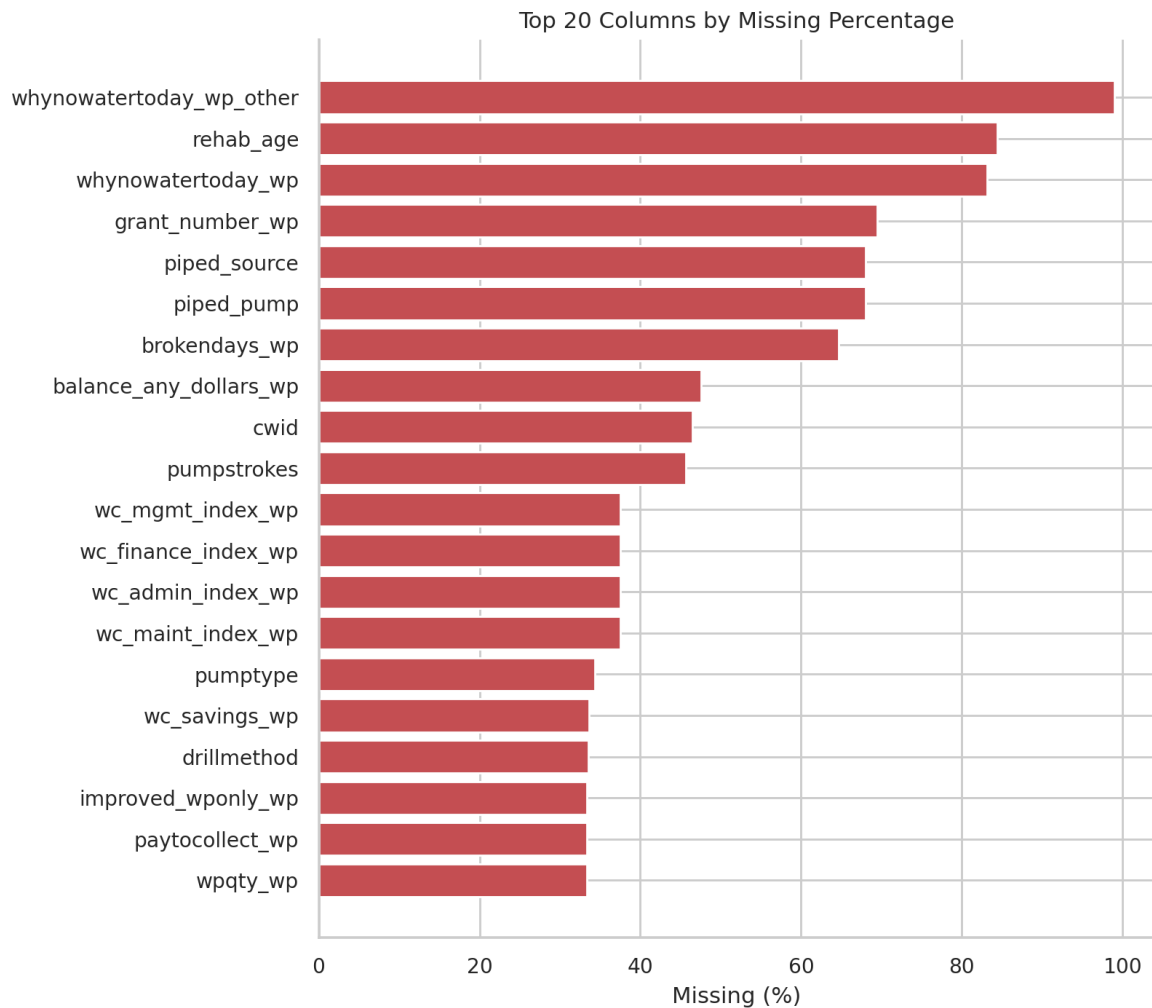


Figure 4.2. Most-missing raw variables (Source: project export)

The top-missing chart includes conditional questions and fields absent in survey blocks. Table 4.1 separates high-volume missingness from its likely mechanism; the model does not assume every blank has the same meaning. This led to Not applicable categories for structurally irrelevant fields and Unknown for genuinely unobserved nominal values.

Table 4.1. Highest missingness among the exported EDA summary

Raw field	Missing	Percent	Interpretation
whynowatertoday_wp_other	1775	99.0%	Review conditional or survey-block context
rehab_age	1513	84.4%	Review conditional or survey-block context
whynowatertoday_wp	1491	83.2%	Review conditional or survey-block context
grant_number_wp	1246	69.5%	Review conditional or survey-block context
piped_pump	1219	68.0%	Review conditional or survey-block context
piped_source	1219	68.0%	Review conditional or survey-block context
brokendays_wp	1160	64.7%	Review conditional or survey-block context

4.4 Duplicate and Invalid-Value Analysis

No exact duplicate rows were found. The `wc_savings_wp` codes 999 and 888 are sentinel values rather than genuine magnitudes.

4.5 Numerical Distributions and Outliers

Usage and age variables are skewed. The 107 recorded negative elevations warrant caution, but observations were not automatically removed or capped.

The distributions in Figure 4.3 and Figure 4.4 show skew and extreme values in age, people served and related counts. The analysis found no basis for blanket deletion; negative elevation values, mainly recorded in India, were retained with a documented quality concern. Table 4.2 connects each observation to treatment.

Table 4.2. Selected numerical extremes and treatment decisions

Variable	Recorded range	IQR flags	Treatment
<code>wp_age</code>	0.5–74	202	Retain; review context
<code>qtypeople_wp</code>	0–5000	99	Retain; review context
<code>qtyhh_wp</code>	0–800	98	Retain; review context
<code>elevation_wp</code>	-217–4493.7	1	Retain; review context
<code>pop_1000</code>	31.7813–9412.56	52	Retain; review context

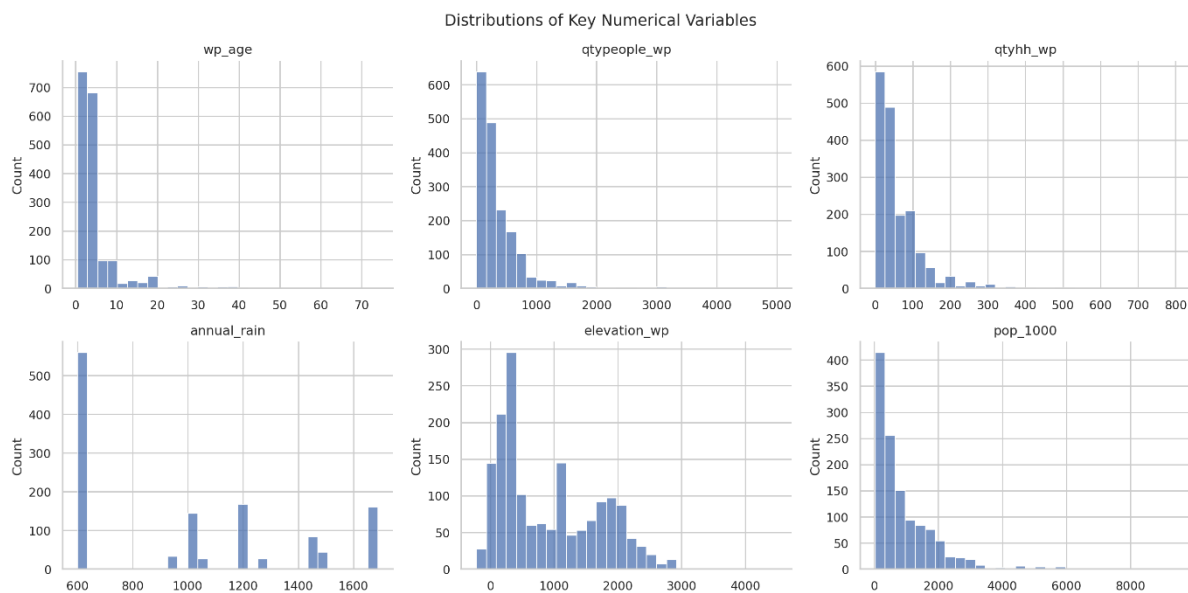


Figure 4.3. Distributions of key numerical predictors before fitted preprocessing

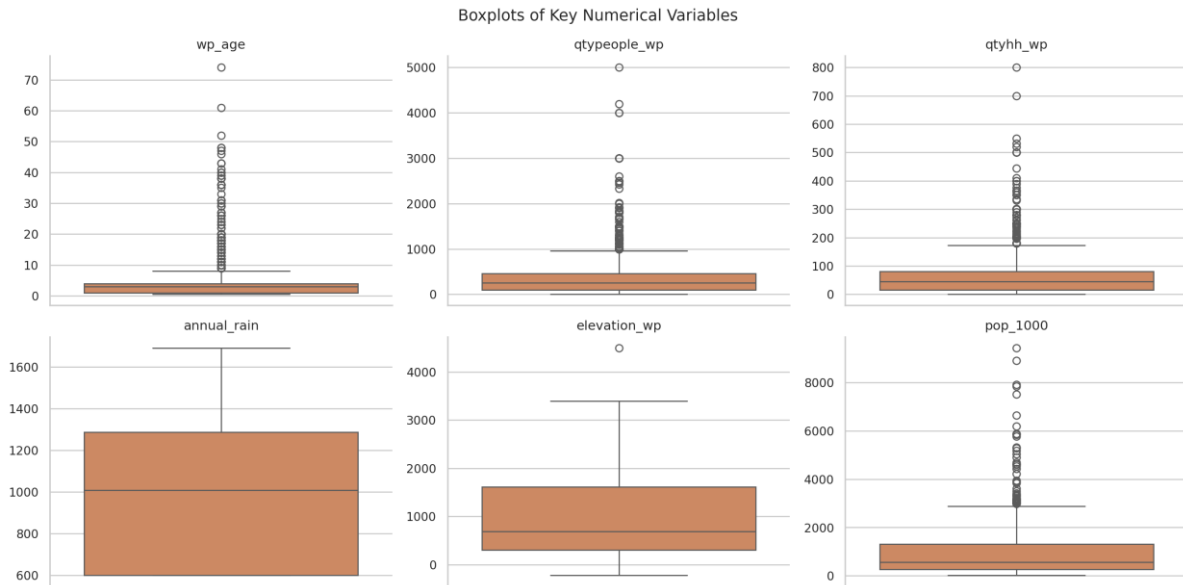


Figure 4.4. Boxplots showing selected numeric extremes retained for contextual review

4.6 Categorical Patterns

Technology, management and geographic categories differ in frequency. Rare categories require a fold-fitted grouping rule.

Figure 4.5 compares selected category levels with target composition. Some rare levels are too small for stable interpretation, so category grouping was fitted within folds rather than chosen from the holdout. This is descriptive association, not evidence that a category causes failure.

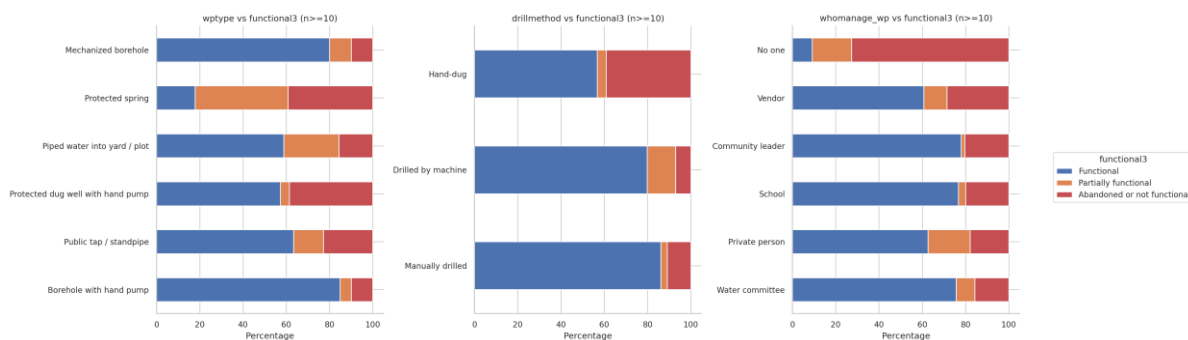


Figure 4.5. Selected categorical levels and their recorded functionality mix

4.7 Functionality by Country

Figure 4.6 shows substantial variation in target composition by country; ordinary stratification cannot establish generalisation to an unseen country.

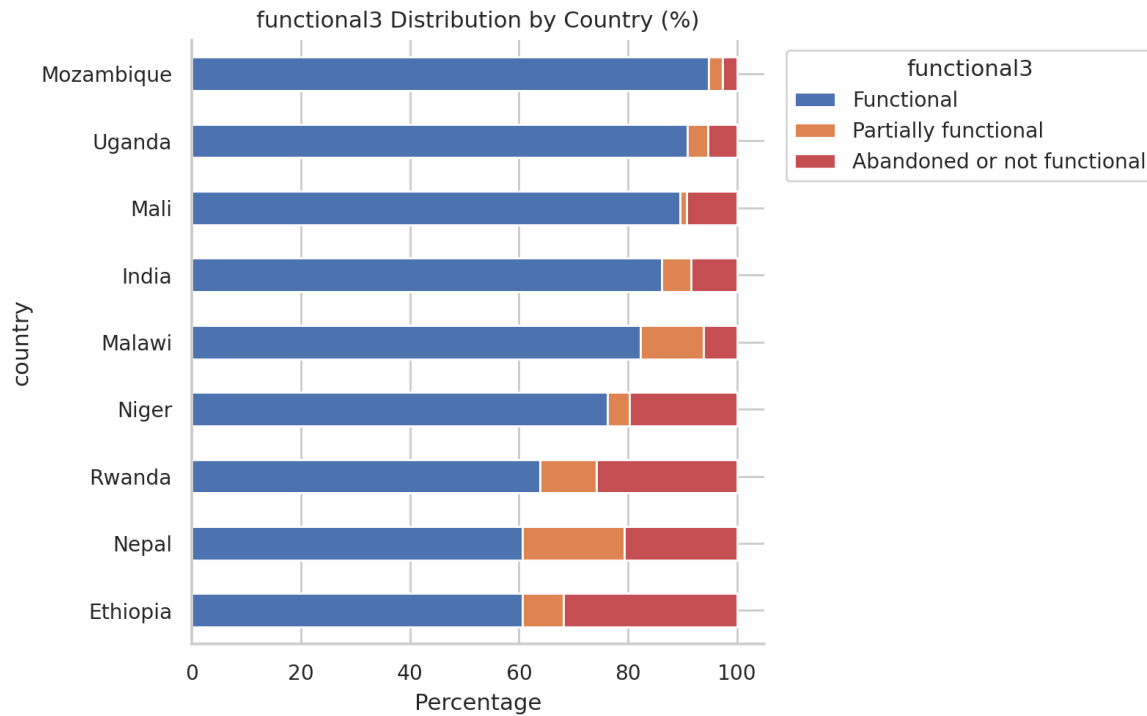


Figure 4.6. Recorded functionality composition by country (Source: project export)

Country composition is visibly heterogeneous: the Functional share ranges from about 60.6% in Ethiopia to 94.7% in Mozambique in the exported EDA table. Because the countries also differ in class support, ordinary stratified CV can benefit from shared country labels without demonstrating transfer to a completely unseen country.

4.8 Numerical Features vs Target

Numerical distributions overlap across classes. Some current-status variables separate classes sharply, signalling leakage rather than useful prospective information.

Figure 4.7 illustrates this overlap and supports the use of a multivariable modelling approach that combines information from multiple features. Variables showing unusually strong class separation were examined carefully, as this could indicate target leakage rather than genuine predictive value. Their inclusion was therefore assessed based on whether the information would realistically be available before determining the water point's functionality.

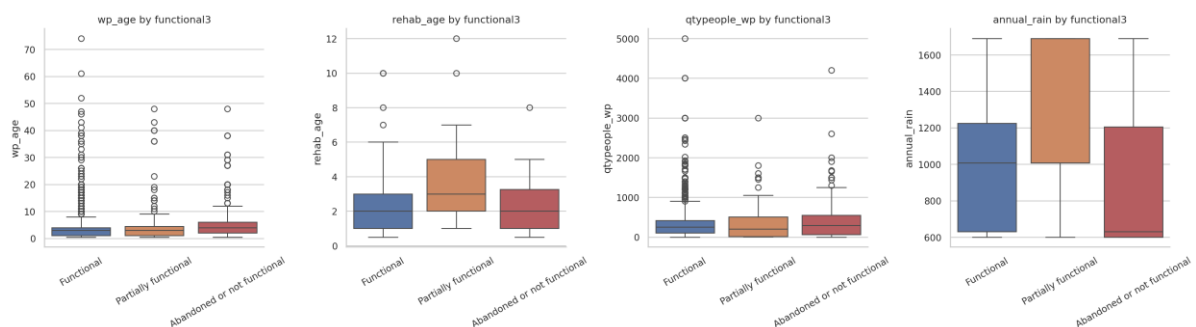


Figure 4.7. Selected numerical predictors compared across the three recorded classes

4.9 Correlation Analysis

Correlation analysis was used as descriptive evidence to identify relationships, redundancy, and possible interactions among numerical variables, rather than as a causal test or a direct rule for feature removal. Figure 4.8 presents these relationships through a correlation heatmap. Highly correlated features were not automatically removed, since tree-based models can use related variables in different ways. Similarly, a strong correlation with the target was interpreted carefully, as it could indicate genuine predictive information or potential target leakage.

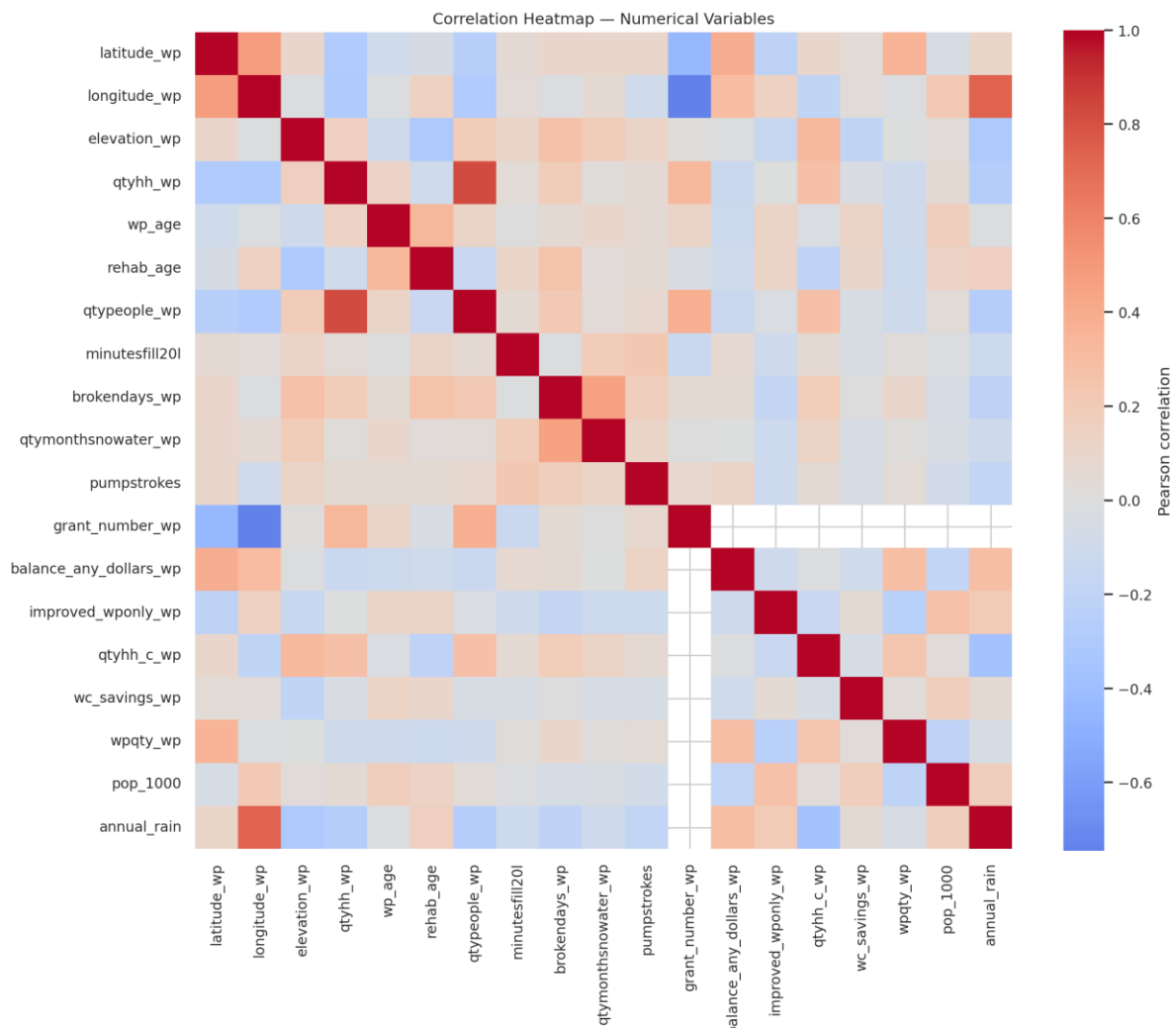


Figure 4.8. Pairwise correlations among numeric survey variables

4.10 Leakage Investigation

Water availability, recent downtime, and related variables were found to strongly reflect the current functionality status, creating a risk of target leakage. These variables were therefore excluded before model training and comparison to ensure that model performance represented genuine predictive ability rather than information derived directly from the target.

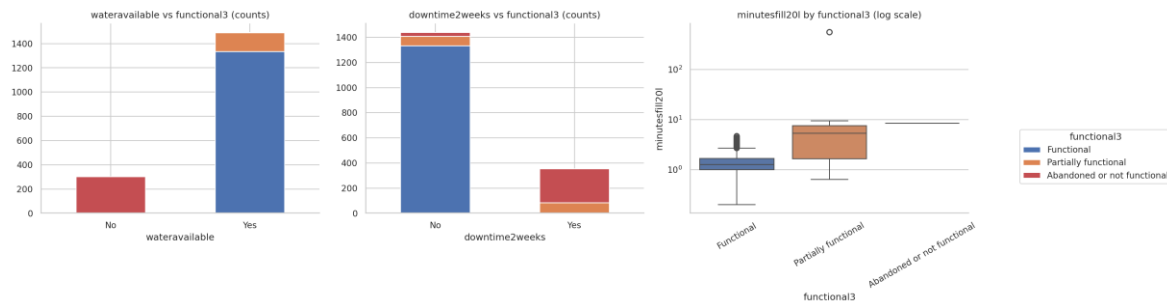


Figure 4.9. Leakage diagnostic for current-status fields (Source: project export)

For example, the EDA crosstab for wateravailable recorded no Functional point among responses of “No”, while the recent-downtime variable strongly separated the recorded classes. Figure 4.9 illustrates the diagnostic. These fields were excluded, despite their apparent predictive strength, because they encode current status.

4.11 Summary of EDA Findings and Resulting Design Decisions

The EDA directly motivated target-aware metrics, semantic missingness treatment, fixed leakage exclusions and country-level robustness checks.

Table 4.3. EDA findings and modelling implications

Finding	Design decision
74.3% majority class	Use Macro-F1 and report per-class performance
Structural and survey missingness	Use semantic categories and missing indicators
Current-status variables reveal target	Exclude before modelling
Country composition differs	Assess cross-country robustness after selection

The EDA decisions feed directly into the modelling contract. The raw class balance determines the metric emphasis and stratified split; high conditional missingness determines explicit missing-value semantics; skewed numeric distributions motivate family-specific transformations; and the leakage audit removes outcome-defining questions. These are design consequences of the evidence, not post-hoc explanations for a preferred estimator.

5. Data Preprocessing

5.1 Leakage-Safe Order of Operations

Rule-based exclusions preceded the split. All learned imputation, encoding, rare-category and feature-selection statistics were fitted within development folds.

Figure 5.1 shows the order that prevents fitted statistics from seeing validation or holdout rows. Fixed semantic exclusions can precede the split; medians, category frequencies, one-hot vocabularies and feature scores must be fitted inside each CV training fold. This order also makes the exported pipeline a faithful inference implementation.

Leakage-safe development sequence

The holdout supplied no fitted statistics during development.

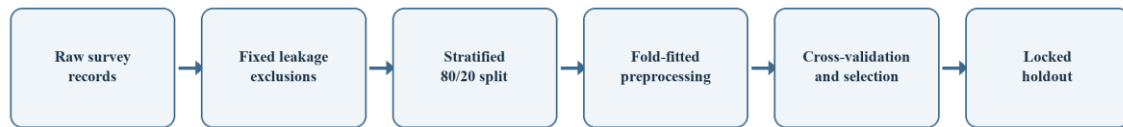


Figure 5.1. Fixed exclusions, stratified split and fold-fitted preprocessing sequence

5.2 Stratified Development/Holdout Split

The fixed random-state-42 split allocated 1,434 records to development and 359 to a historical holdout. The holdout was opened only after final configuration lock.

The 80:20 split produced 1,434 development records and 359 historical holdout records. Within development, repeated stratified CV was used for comparison and tuning. The holdout was held apart from estimating imputers, category vocabularies, feature scores and model parameters. This separation is especially important with a small minority class: consulting holdout errors during design would quietly turn the final check into another tuning set.

5.3 Missing-Value Treatment

Conditional fields such as pump or rehabilitation attributes receive explicit Not applicable semantics where appropriate; other nominal unknowns use Unknown. Numeric fields receive fold-fitted imputation and selected missing indicators.

Missing values were treated according to their meaning rather than using a single method for all features. For example, a missing pump-type value for a water point without a pump represents “Not Applicable,” while a missing country or other expected value represents “Unknown.” Numerical missing values were generally handled using median imputation, with missing-value indicators added where the absence of data could itself provide useful information. Table 5.1 summarises these treatment categories.

Table 5.1. Missing-value semantics in the production pipeline

Situation	Example	Treatment
Structurally irrelevant nominal	Pump or piped attributes	Not applicable category
Unobserved nominal	Country / management field	Unknown category
Structural numeric	rehab_age	Zero with missing flag
Other numeric gaps	Selected usage / environment fields	Training-fold median and indicators
Sentinel-coded savings	wc_savings_wp = 999 or 888	Convert to missing before fitted steps

5.4 Sentinel and Invalid-Value Handling

The FixedDataCleaner handles special missing-value codes within the production preprocessing pipeline by converting 999 and 888 in wc_savings_wp to missing values. This ensures that these codes are treated as unavailable data rather than genuine numerical values before further preprocessing is applied.

5.5 Categorical Encoding and Rare Categories

Nominal variables are one-hot encoded, with infrequent levels grouped at the final frequency threshold of 10. Unknown categories are tolerated at inference.

Nominal categories have no defensible ranking, so one-hot encoding avoids imposing one. Rare-category grouping curbs nearly empty columns; its frequency threshold is learned from training rows only. The threshold was later compared at 5, 10 and 20 on development data, with 10 retained.

5.6 Ordinal Encoding

Committee indices were encoded as ordered categories to preserve their natural progression: Inadequate < Minimum < Moderate < Advanced. Missing values were handled separately to avoid incorrectly assigning them a position within this ordered scale.

5.7 Scaling and Log Transform Strategy

Scale-sensitive models received their own numeric representation. Tree models used a base representation without forcing a common scaler across all families.

5.8 Model-Family-Specific Preprocessing

Logistic Regression used a scaled representation with skew treatment because it is sensitive to feature scale. Tree-based models used a tree-oriented representation, while CatBoost was also evaluated using its native handling of categorical features.

Using model-specific preprocessing avoided forcing the same transformation onto algorithms with different requirements. This allowed each model family to be evaluated with a suitable practical pipeline, resulting in a fairer comparison.

5.9 Preprocessing Pipeline and Reproducibility

The serialized production pipeline retains deterministic cleaning, fitted preprocessing, selection and classification. Its exported metadata records the input schema and feature counts.

Table 5.2. Preprocessing by feature type

Feature type	Production treatment
Nominal	Semantic missing categories; rare grouping; one-hot encoding
Committee indices	Ordered encoding and separate missing treatment
Skewed numerical	Fold-fitted imputation and tree-oriented base treatment
Sentinel-coded savings	999/888 to missing before learned transformations

Figure 5.2 traces the deployed transform from 32 named raw columns through 102 processed columns and 60 selected columns to the 336-tree RF. The exported artifact retains all fitted components, preventing the API from recreating training-time preprocessing from source code alone.

Locked inference representation

The same fitted transformations are serialized with the classifier.



Figure 5.2. 32 raw predictors transformed to 102 columns and 60 selected RF features

6. Feature Engineering and Feature Selection

6.1 Engineered Features Evaluated

The development workflow assessed additional semantic indicators and derived quantities, including usage or age relationships, without altering the target.

The feature engineer creates two fixed missingness flags and four optional derived variables. Optional features were tested rather than assumed useful. Table 6.1 records each definition and whether the final base representation includes it; “base” still retains the fixed semantic flags.

Table 6.1. Engineered feature definitions and final inclusion

Feature	Definition or intent	Final base model
rehab_age_missing	Flag absent rehabilitation age	Included semantic flag
pop_1000_missing	Flag absent surrounding population	Included semantic flag
wp_age_group	Training-fitted age groups	Optional; not used
was_rehabilitated	Yes/No numerical mirror	Optional; not used
people_per_household	People divided by households, guarded	Optional; not used
has_management_committee	Water-committee indicator	Optional; not used

6.2 Base vs Engineered Feature Sets

Figure 6.1 compares feature representations by family. The selected Random Forest retained the base tree representation because the engineered variant did not justify replacing it.

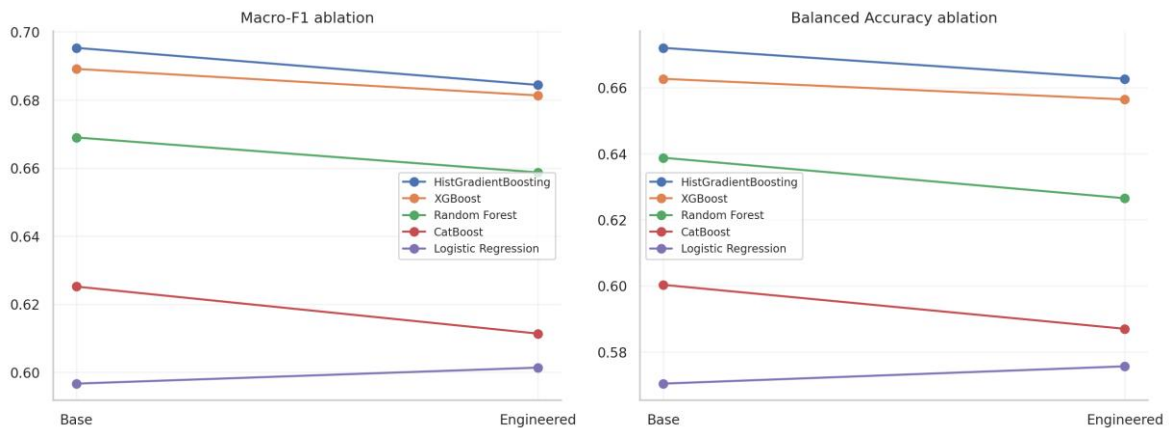


Figure 6.1. Base versus engineered feature ablation (Source: project export)

The same repeated-CV folds compared base and engineered variants. Four tree-family winners declined by roughly 0.008–0.014 Macro-F1 with the optional additions, whereas Logistic Regression gained about 0.0047. Table 6.2 quantifies the ablation; Figure 6.1 plots it. The final RF therefore retained its base representation.

Table 6.2. Repeated-CV base versus engineered Macro-F1 by family

Family	Base	Engineered	Difference
HistGradientBoosting	0.6953	0.6844	-0.0109
XGBoost	0.6891	0.6813	-0.0078
Random Forest	0.6689	0.6587	-0.0103
CatBoost	0.6251	0.6113	-0.0138

Logistic Regression	0.5966	0.6013	+0.0047
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6.3 Feature-Selection Method

Mutual-information SelectKBest was fitted inside development CV to rank processed features without using the holdout.

6.4 SelectKBest Experiments

The accepted k=60 experiment gave exploratory five-fold Macro-F1 around 0.7099; final repeated CV was used for selection.

A development-only five-fold experiment favoured k=60 over k=80, but that exploratory score was not directly substituted for final repeated CV. The accepted configuration was re-evaluated alongside other finalists with the common repeated protocol. Table 6.3 separates the exploratory comparison from the final selection result.

Table 6.3. Development-only mutual-information feature-selection comparison

Selected k	Exploratory Macro-F1	Balanced accuracy	Partial F1
60	0.7099	0.6792	0.4330
80	0.6920	0.6618	0.4188

6.5 Final Production Representation

The production preprocessor outputs 102 columns before selection; SelectKBest retains 60. These are processed columns derived from 32 raw inputs.

Feature selection operates after fitted preprocessing, where a raw categorical field can expand into several columns. Thus 60 selected columns do not mean 60 new survey questions: the deployed request still requires 32 named raw fields. The selector and column order are serialized with the model, so inference uses the exact same transformation learned from development records.

6.6 Final 32-Input Schema

Appendix B lists the exact schema names from the locked model metadata.

7. Modelling Methodology

7.1 Experimental Design

Stage 6 compared five model families. Stage 7 used development-only optimisation, then selected a final configuration before the one-time holdout evaluation.

7.2 Validation Strategy

The main baseline and finalist comparisons used RepeatedStratifiedKFold with five folds and three repeats. Training and validation results were retained to diagnose overfitting.

Each baseline family used five stratified folds repeated three times. Out-of-fold predictions supported confusion matrices and class-level diagnostic plots, while training scores exposed gaps. The 359-row holdout stayed separate until the development evidence selected the RF. Table 7.1 makes these evidence sources distinct.

Table 7.1. Evidence stages and holdout access

Stage	Rows / validation	Purpose	Holdout access
Baseline	1,434; 5 folds x 3 repeats	Five-family comparison	None
Search	1,434; 5-fold search	Tune candidates	None
Finalist	1,434; 5 folds x 3 repeats	Lock final RF	None
Historical check	359 once	Report generalisation	After lock
Post-hoc	Development diagnostics	Interpret robustness	No replacement selection

7.3 Evaluation Metrics

Macro-F1 was the primary criterion. Balanced accuracy, macro precision, MCC, weighted F1, accuracy, log loss, average precision, class metrics and train–validation gap added context.

Macro-F1 averages the three class F1 values, giving the scarce Partial class the same weight as Functional. Recall answers how many actual members of a class are found; precision answers how many predicted members are correct. Their harmonic mean penalises a model that improves one while sacrificing the other. Balanced accuracy averages class recalls, while MCC considers all confusion-matrix cells. Taken together, these measures make the selection criterion and its operational weaknesses visible.

The report treats log loss and average precision as supplementary probability and ranking diagnostics. The displayed maximum probability is not a calibrated confidence interval. A favourable ranking score can coexist with poor classification at the default decision rule, so the three-class confusion matrix and per-class support remain necessary for interpreting the final result.

7.4 Candidate Model Families and Rationale

Logistic Regression provided a linear reference. Random Forest supplied variance-reduced nonlinear trees; HistGradientBoosting, XGBoost and CatBoost supplied boosted alternatives.

The five families test different inductive assumptions on the same leakage-safe input contract. Logistic Regression supplies a linear reference after scale-sensitive preprocessing. Bagged Random Forest trees can represent interactions without a single global linear relationship. Histogram boosting, XGBoost and CatBoost are alternative additive-tree approaches, with CatBoost also tested through a categorical-aware route. Their scores compare complete fitted pipelines rather than isolated estimator names.

7.5 Baseline Configurations

The baseline family winner for each family was chosen from its own recorded preprocessing/feature-set variants rather than a single shared representation.

The baseline stage selected the strongest recorded representation within each family before family-level comparison. This matters because one-hot expansion, scaling and optional derived variables affect estimators differently. A single shared preprocessing recipe would have confounded a weak representation with a weak model. The baseline table therefore identifies the family and its selected feature treatment together.

7.6 Hyperparameter Optimization Strategy

RandomizedSearchCV searched candidate parameter sets using development data only. Search budgets were 30 HistGB, 35 XGBoost, 30 RF and 12 CatBoost iterations.

RandomizedSearchCV explored bounded parameter combinations rather than exhaustively enumerating a grid. The recorded budgets were 30 HistGB, 35 XGBoost, 30 RF and 12 CatBoost draws; each used five folds. Search-best scores are informative but are not interchangeable with the later finalist repeated-CV estimates.

7.7 Class-Imbalance Strategies

Class weighting and resampling were explored. SMOTENC required a safe mixed representation and was investigated later as a post-hoc control, not as a reason to rewrite the original lock.

Minority-class interventions were judged against the complete error profile. Weighting can increase the cost of missing Partial cases, and oversampling can increase their presence in a training fold, but either can produce more false alarms or reduce overall Macro-F1. Resampling was fitted only inside training partitions. The later SMOTENC comparison used a valid mixed numeric and categorical representation and was kept outside the original selection decision.

7.8 Final Evaluation Protocol

The final selection record explicitly says `holdout_evaluated=false`. The 359-record historical holdout was evaluated once afterward; no post-holdout tuning was recorded.

The final decision sequence is an important control on optimism. Development-only experiments determined preprocessing, rare-category threshold, selected-feature count and RF configuration. The lock then recorded a model and expected schema before the 359-row holdout was opened for a single evaluation. Subsequent feature-importance and country-transfer checks describe the locked choice; they do not retroactively change it.

Experimental evaluation protocol

The historical holdout was evaluated once after the final selection.

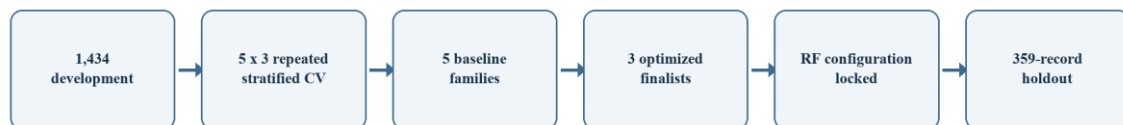


Figure 7.1. Development-only model selection followed by the one-time historical holdout

8. Results, Optimization and Final Model Selection

8.1 Baseline Model Comparison: All Five Families

Table 8.1. Stage-6 baseline family winners

Family	Winning variant	Macro-F1 $\pm$ SD	MCC
HistGradientBoosting	HistGB Base	0.6953 $\pm$ 0.0274	0.6332
XGBoost	XGBoost Base	0.6891 $\pm$ 0.0283	0.6285
Random Forest	RF Base	0.6689 $\pm$ 0.0288	0.6128
CatBoost	CatBoost Base	0.6251 $\pm$ 0.0326	0.5592
Logistic Regression	LR Engineered	0.6013 $\pm$ 0.0402	0.5492

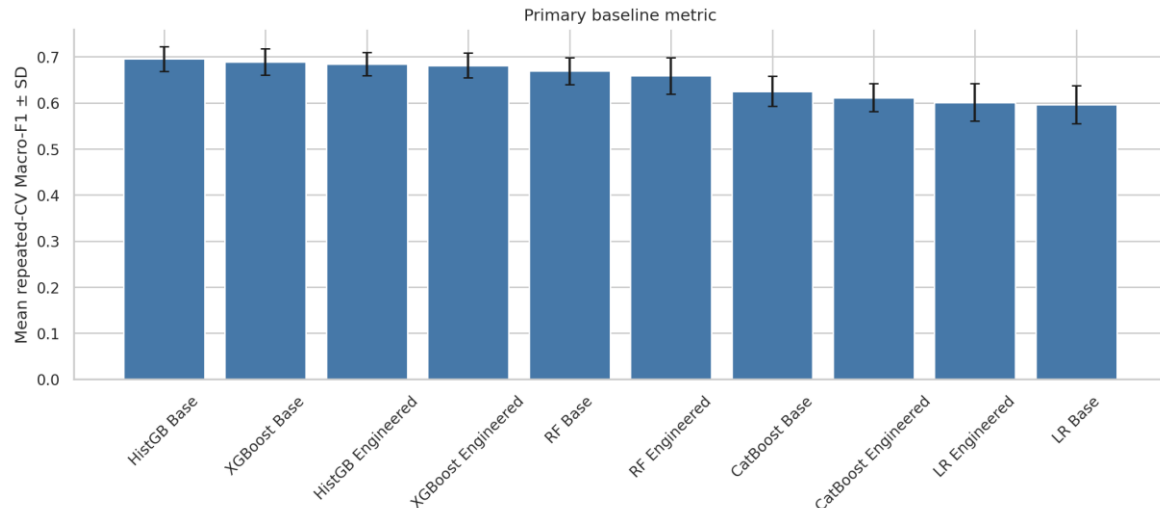


Figure 8.1. Baseline Macro-F1 by model family (Source: project export)

8.2 Base vs Engineered Ablation

Base tree variants won for the strongest tree families; Logistic Regression preferred the engineered scaled variant.

The base/engineered ablation has practical value because derived features can add noise or unstable ratios in a small survey. The RF base winner scored 0.6689 Macro-F1 versus 0.6587 engineered. The final selected representation combined this base choice with later fold-fitted feature selection.

8.3 Baseline Error and Minority-Class Analysis

Baseline Partial-class recall and F1 were low across families, revealing a persistent minority-class challenge.

Figure 8.2 shows that strong training fit did not guarantee validation quality. The Partial class was weak for every family, as the saved OOF metrics show. These errors motivated a shortlist based on Macro-F1 and supporting class metrics rather than on training score.

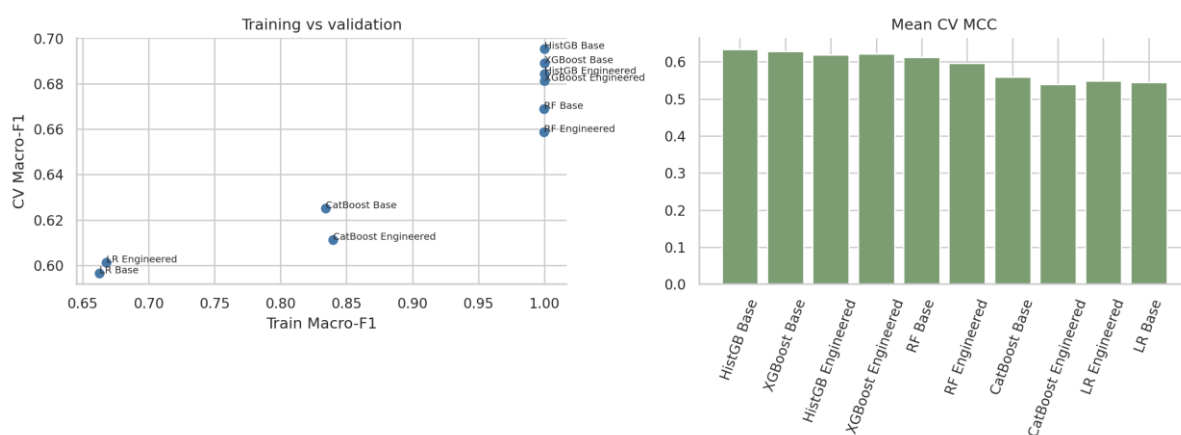


Figure 8.2. Stage-6 training-versus-validation Macro-F1 and MCC diagnostics

8.4 Shortlist Selection

HistGradientBoosting, XGBoost and Random Forest advanced because of their development-only baseline evidence.

8.5 Hyperparameter Search Results

Search best Macro-F1 values were approximately 0.6945 HistGB, 0.6920 XGBoost, 0.6975 RF and 0.6494 CatBoost. The search result alone was not the final selection.

Table 8.2 gives the best search scores and budgets. RF had the highest recorded search Macro-F1 at 0.6975, while CatBoost's challenger remained lower at 0.6494. The final decision waited for the common repeated-CV finalist comparison, avoiding selection from incomparable search summaries.

Table 8.2. RandomizedSearchCV best scores and search budgets

Family	Draws	Fits	Best Macro-F1
HistGradientBoosting	30	150	0.6945
XGBoost	35	175	0.6920
Random Forest	30	150	0.6975
CatBoost	12	60	0.6494

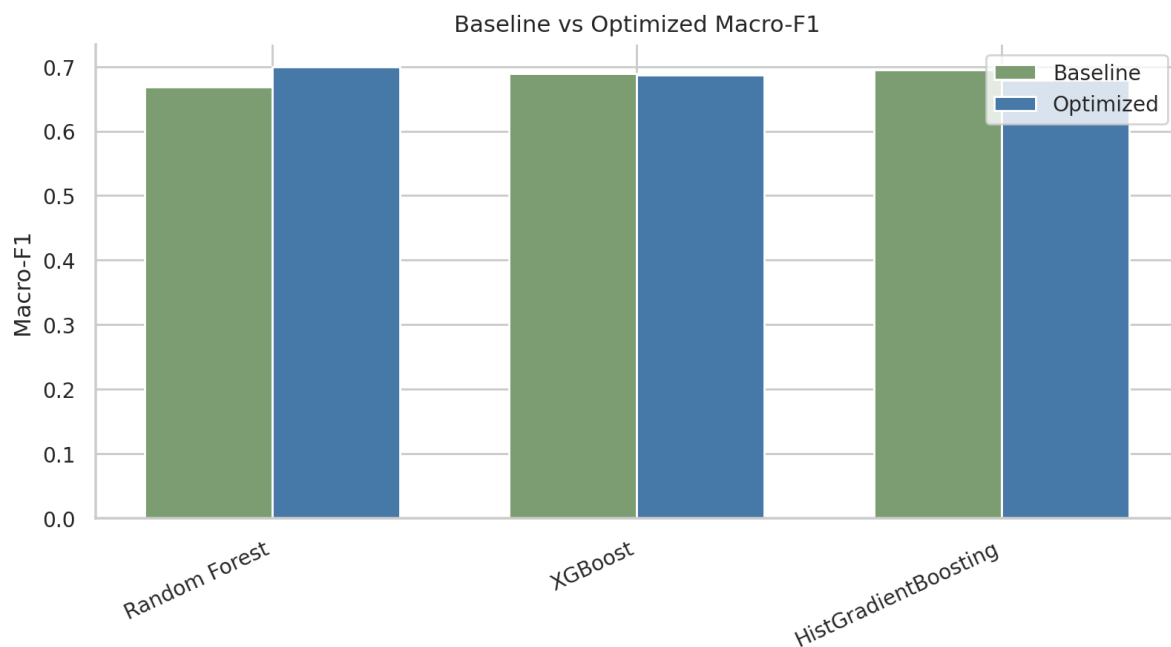


Figure 8.3. Baseline and optimized Macro-F1 across shortlisted families

8.6 Imbalance-Treatment Experiments

Alternative weighting and resampling strategies were considered against Macro-F1 and minority-class trade-offs. The selected configuration retained no imbalance treatment.

Class weighting and RandomOverSampler raised some Partial recall values but could reduce precision, MCC or Macro-F1. For the RF, balanced weighting raised Partial recall to 0.5000 while Macro-F1 fell to 0.6873; ordinary random oversampling produced 0.4286 Partial recall and 0.6953 Macro-F1. The unweighted path remained stronger on the primary criterion.

Table 8.3. Random Forest imbalance-treatment trade-offs on development evidence

Treatment	Macro-F1	Partial recall	MCC
none	0.6985	0.3333	0.6200
balanced	0.6873	0.5000	0.5921

balanced_subsample	0.6873	0.5000	0.5921
ros	0.6953	0.4286	0.6096

8.7 Rare-Category Threshold Experiments

A frequency threshold of 10 was accepted for the final tree representation.

Development-only rare thresholds 5, 10 and 20 produced Macro-F1 values 0.6908, 0.6985 and 0.6963. Table 8.4 shows the near comparison. Ten was retained because it led this experiment while limiting category fragmentation; the holdout did not participate.

Table 8.4. Rare-category threshold comparison on development data

Minimum frequency	Macro-F1	Balanced accuracy	Partial F1
5	0.6908	0.6604	0.4211
10	0.6985	0.6652	0.4421
20	0.6963	0.6657	0.4352

8.8 Feature-Selection Results

SelectKBest(k=60) improved the Random Forest development evidence; the final repeated-CV comparison determined the lock.

8.9 Finalist Repeated-CV Comparison

Table 8.5 compares the three optimisation finalists under repeated CV; the Random Forest led on Macro-F1.

Table 8.5. Finalist repeated-CV results

Finalist	Macro-F1 $\pm$ SD	Balanced accuracy	MCC
Random Forest	0.6996 $\pm$ 0.0481	0.6691	0.6349
XGBoost	0.6874 $\pm$ 0.0291	0.6594	0.6264
HistGradientBoosting	0.6793 $\pm$ 0.0221	0.6497	0.6216

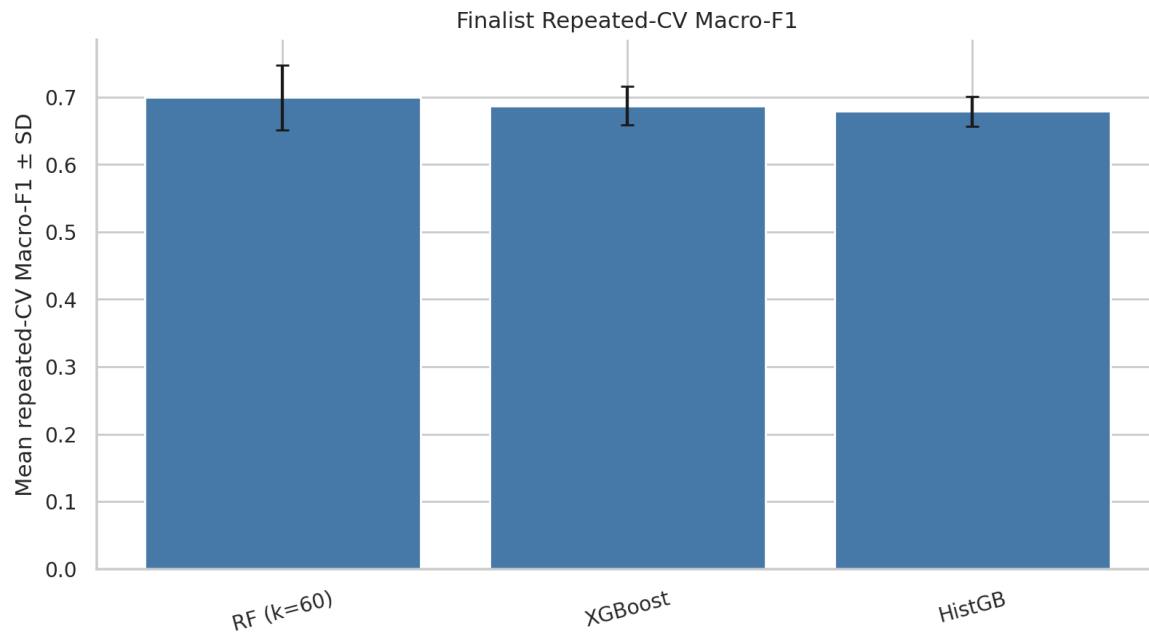


Figure 8.4. Finalist repeated-CV Macro-F1 (Source: project export)

As Figure 8.4 shows, the selected RF scored 0.6996 ± 0.0481 Macro-F1, compared with 0.6874 for XGBoost and 0.6793 for HistGB. The supporting RF MCC of 0.6349 and smaller train–CV gap reinforced the choice.

8.10 Final Model Selection and Justification

The selected RF used 336 trees, `max_features=0.5`, `min_samples_split=8`, `min_samples_leaf=5`, `bootstrap=False` and no max-depth limit. It was locked before holdout evaluation.

Relative to the Stage-6 RF family winner, final repeated-CV Macro-F1 improved by 0.0306, MCC by 0.0222, and the train–CV gap fell from 0.3307 to 0.1652. Table 8.6 summarises the comparison. The model lock recorded these development results before the one-time holdout test.

Table 8.6. Random Forest baseline to locked repeated-CV change

Metric	Stage-6 baseline	Locked RF	Change
Macro-F1	0.6689	0.6996	+0.0306
MCC	0.6128	0.6349	+0.0222
Train–CV gap	0.3307	0.1652	-0.1655

8.11 Historical Holdout Performance

The one-time holdout achieved Macro-F1 0.6699, balanced accuracy 0.6392, MCC 0.6466 and accuracy 0.8663. The accuracy must be read alongside the class imbalance.

Table 8.7 provides the complete stored historical holdout summary. Figure 8.5 juxtaposes the development estimate and the one-time check; the lower holdout Macro-F1 is consistent with uncertainty in a small, imbalanced sample. No post-holdout retuning followed.

Table 8.7. One-time historical holdout metrics for the locked RF

Metric	Recorded value	Interpretive role
Macro-F1	0.6699	Equal class weight
Balanced accuracy	0.6392	Average class recall
MCC	0.6466	Confusion-matrix association
Weighted F1	0.8462	Support-weighted class fit
Accuracy	0.8663	Overall correct fraction
Log loss	0.3353	Probability quality indicator
Macro AP	0.7930	Class ranking quality

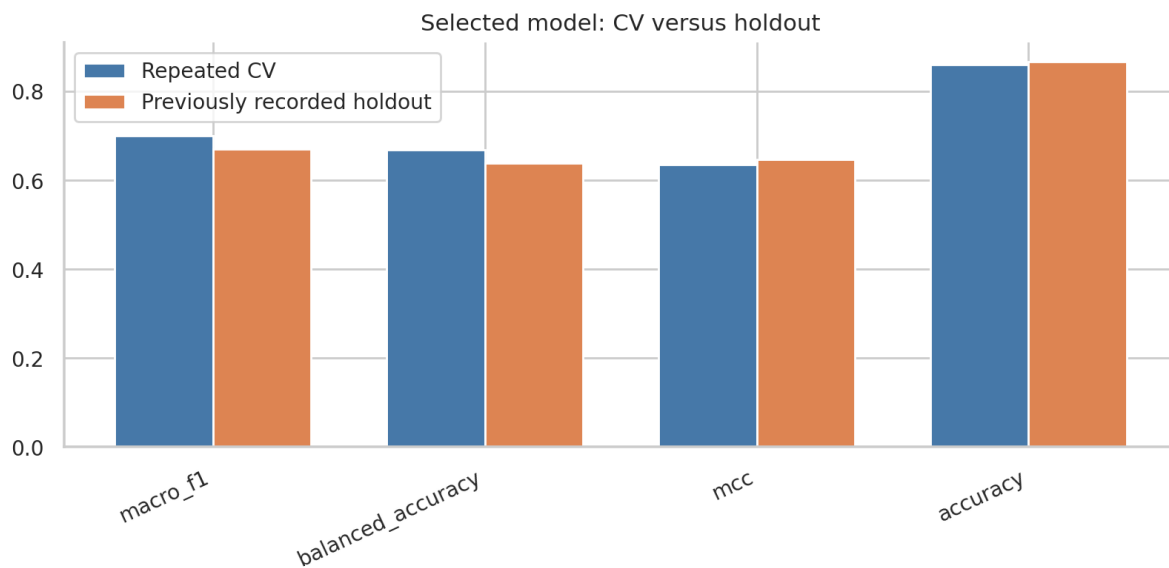


Figure 8.5. Locked RF repeated-CV metrics versus the previously recorded holdout

8.12 Per-Class Holdout Performance

Functional F1 was 0.9234 and Abandoned F1 was 0.7863. Partial F1 was only 0.3000, with recall 0.1935.

Table 8.8. Historical holdout per-class performance

Class	Support	Precision	Recall	F1
Functional	267	0.8810	0.9700	0.9234
Partially functional	31	0.6667	0.1935	0.3000
Abandoned or not functional	61	0.8214	0.7541	0.7863

The high Functional recall of 0.9700 masks the Partial recall of 0.1935. Although weighted F1 was 0.8462, it is dominated by the majority class. Average precision for Partial was 0.5208, providing a ranking perspective but not solving the low thresholded recall.

8.13 Confusion Matrix Analysis

Among the 31 actual Partially Functional records, only six were correctly classified. Of the remaining cases, 22 were incorrectly predicted as Functional and three as Abandoned/Not Functional. This represents the model's main operational weakness, as most errors involve partially functional water points being mistaken for fully functional ones.

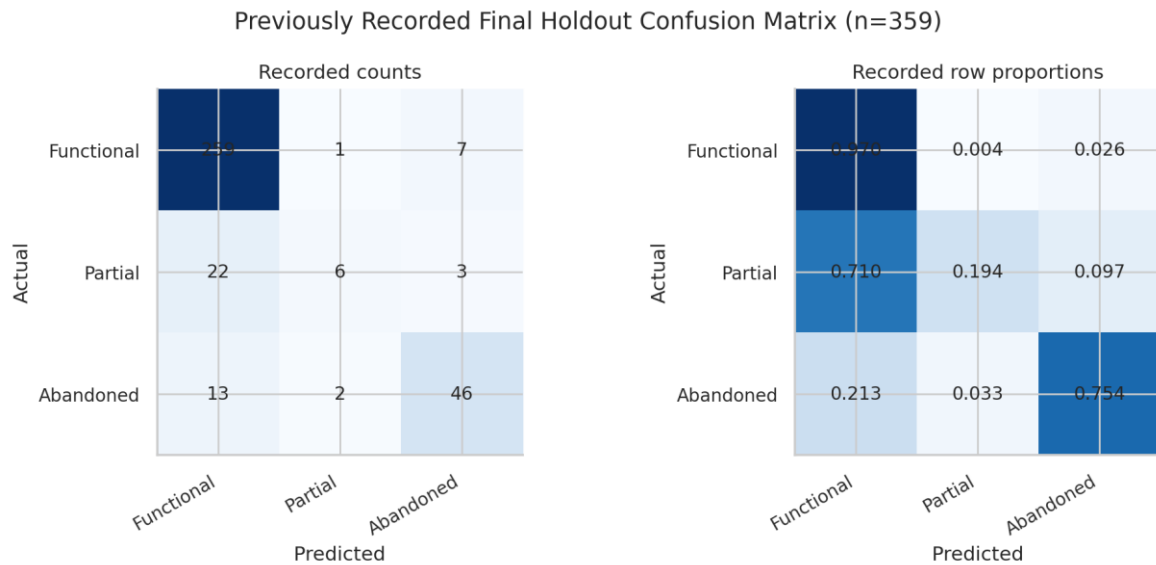


Figure 8.6. Historical holdout confusion matrix (Source: project export)

The holdout confusion matrix contains 22 Partial-to-Functional errors among only 31 actual Partial cases. Operationally, that error can direct attention away from degrading infrastructure. The interface therefore shows all class probabilities and cautions users against treating the label as a definitive inspection outcome.

8.14 Summary of Results

The final RF led the development comparison and passed the one-time historical check, but Partial recognition and cross-country transfer remain material limits.

The results support the locked RF as the best of the tested finalists under the stated development criterion. They do not establish that RF is universally best for water-point surveys. The historical holdout is lower than repeated CV, and the Partial confusion pattern remains a material weakness. This conclusion preserves the distinction between selecting a reproducible candidate, evaluating it once, and deciding whether it is fit for a particular field workflow.

9. Model Robustness and Interpretation

9.1 Purpose and Post-Hoc Status

Notebook 06 was completed after the Stage-7 model lock. Its diagnostics probe reliability; they were not used for the original selection.

9.2 Raw Permutation Importance

Permutation results highlighted pumpstrokes and several geographic fields. Importance measures predictive reliance in this dataset, not causal impact.

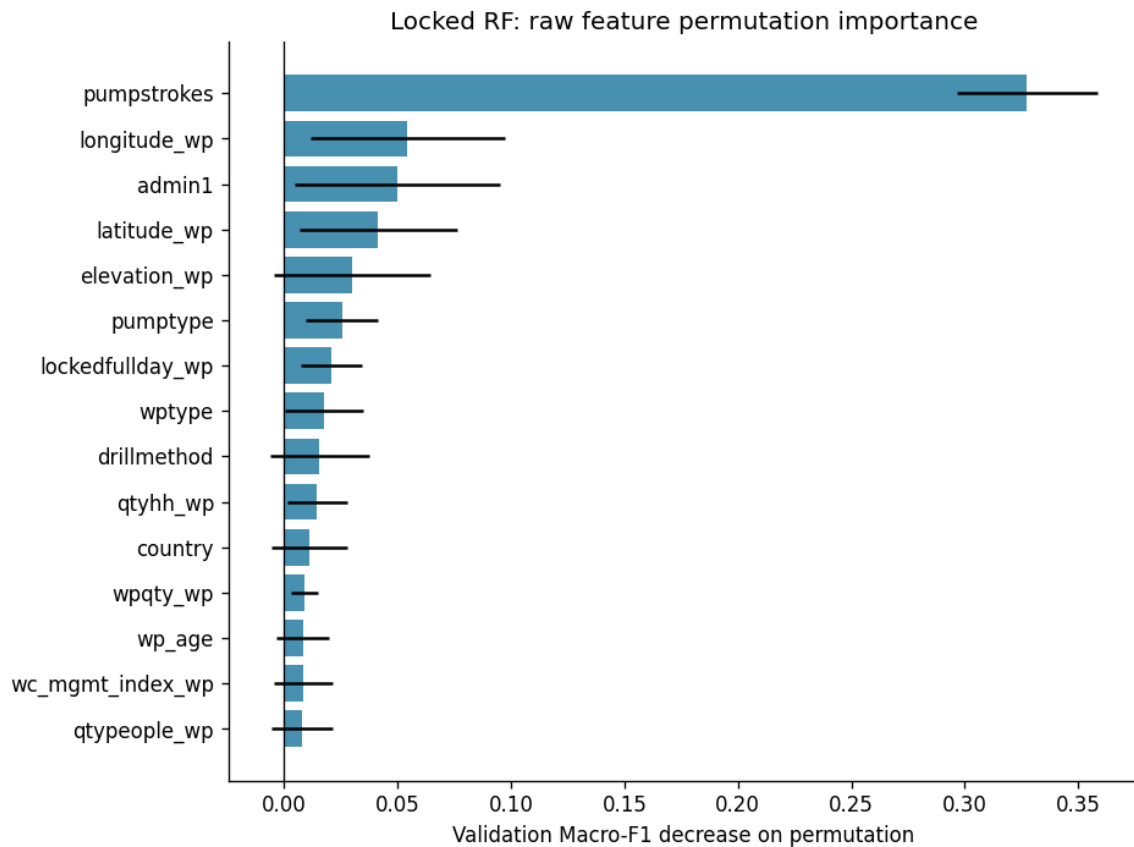


Figure 9.1. Raw-feature permutation importance (Source: project export)

Figure 9.1 places pumpstrokes highest among the saved raw permutation scores, followed by geographic attributes such as longitude, admin1, latitude and elevation. Permuting a correlated predictor can understate or redistribute reliance; the ranking is descriptive of this fitted model and dataset.

9.3 Selected RF Feature Importance

The processed-feature view describes how the selected 60 columns contribute inside the RF; correlated variables can share or distort attribution.

Figure 9.2 shows importance across the selected processed columns. Because one raw categorical field can expand into several one-hot features, this view answers a different question from raw permutation importance. Neither plot supports a causal statement about failure mechanisms.

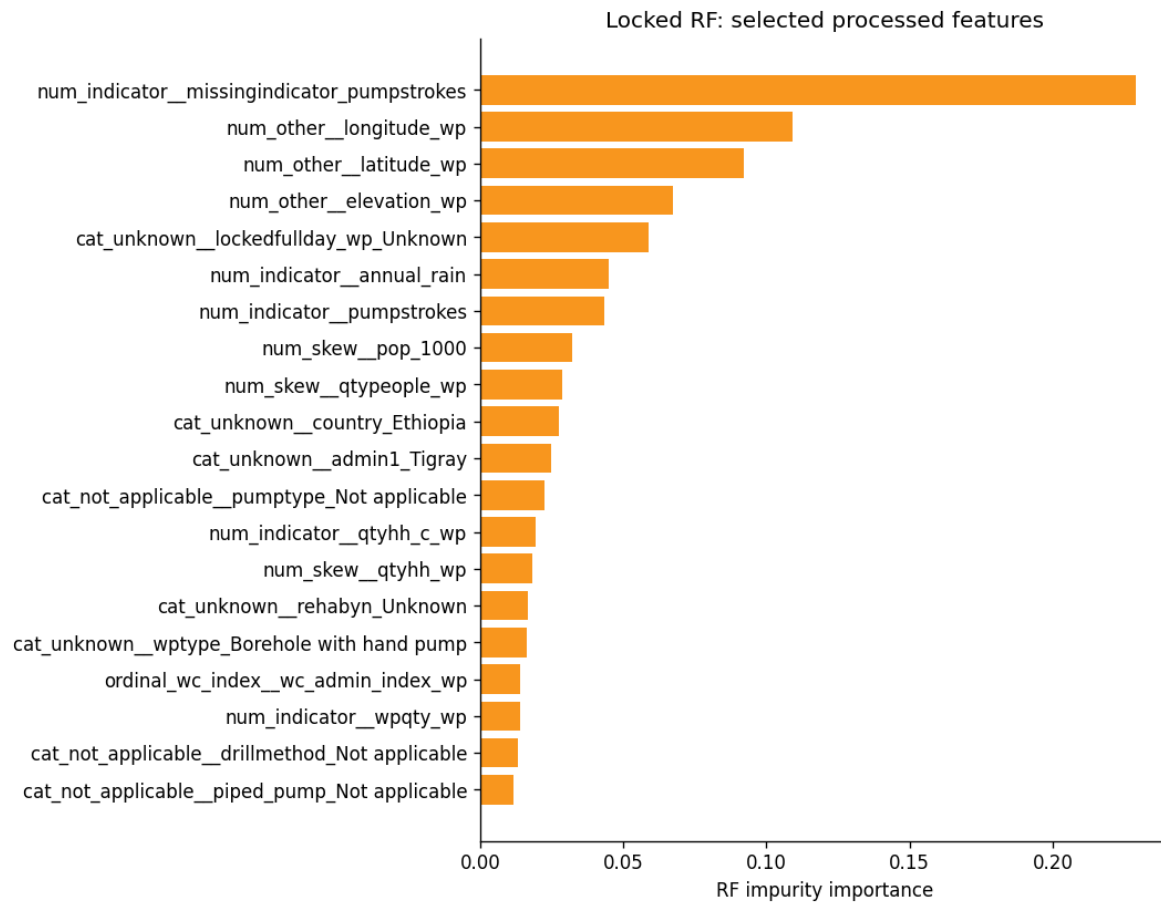


Figure 9.2. Importance of selected processed RF features in the locked model

9.4 Coordinate Ablation

Removing coordinates reduced repeated-CV Macro-F1 from 0.6996 to 0.6848, indicating useful geographic signal but raising transfer concerns.

Table 9.1 compares the locked RF with three post-hoc controls. Coordinate removal reduced Macro-F1 to 0.6848. This supports the value of location for within-sample prediction while sharpening the question of geographic transfer.

Table 9.1. Post-hoc control comparison; not original model selection

Post-hoc configuration	Macro-F1	Partial recall	MCC
Locked RF	0.6996	0.3205	0.6349
RF without coordinates	0.6848	0.2808	0.6163
Mixed representation control	0.6300	0.3072	0.4970
SMOTENC RF	0.6456	0.4285	0.5084

9.5 Representation Control

A mixed-representation Random Forest was also evaluated as a control experiment, achieving a Macro-F1 score of approximately 0.6300. Since this was lower than the performance of the locked final model, it did not provide sufficient evidence to replace the selected pipeline.

9.6 Safe SMOTENC Experiment

The post-hoc SMOTENC RF improved Partial recall to about 0.4285 while reducing overall Macro-F1 to about 0.6456. It did not justify replacement.

SMOTENC was tested only after the lock using a valid mixed representation. It raised Partial recall from about 0.3205 to 0.4285 in the post-hoc comparison, but Macro-F1 fell to 0.6456 and MCC to 0.5084. That trade-off did not justify replacing the official model.

The later SMOTENC control used the mixed numeric/categorical resampling approach described by imbalanced-learn [11]. It remains explicitly post-hoc.

9.7 Learning-Curve Analysis

Learning curves and the 0.1652 train-CV Macro-F1 gap indicate remaining generalisation limits.

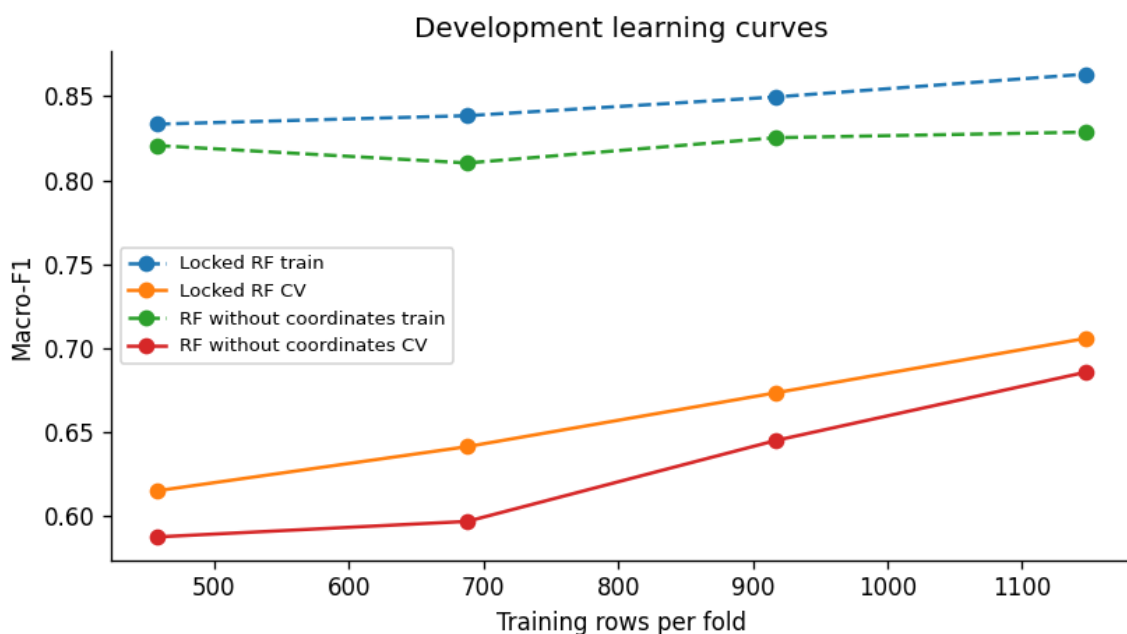


Figure 9.3. Post-hoc learning curves (Source: project export)

The learning curves compare training and validation Macro-F1 as development sample size grows. Their gap indicates that additional records could help, especially if they cover the minority class and underrepresented contexts. The curves are diagnostic summaries over existing development data; they are not a forecast of exactly how much a new survey would improve the model.

9.8 Cross-Country Robustness

Country-level and leave-one-country-out diagnostics were weaker than ordinary stratified CV, so transfer to unseen countries is uncertain.

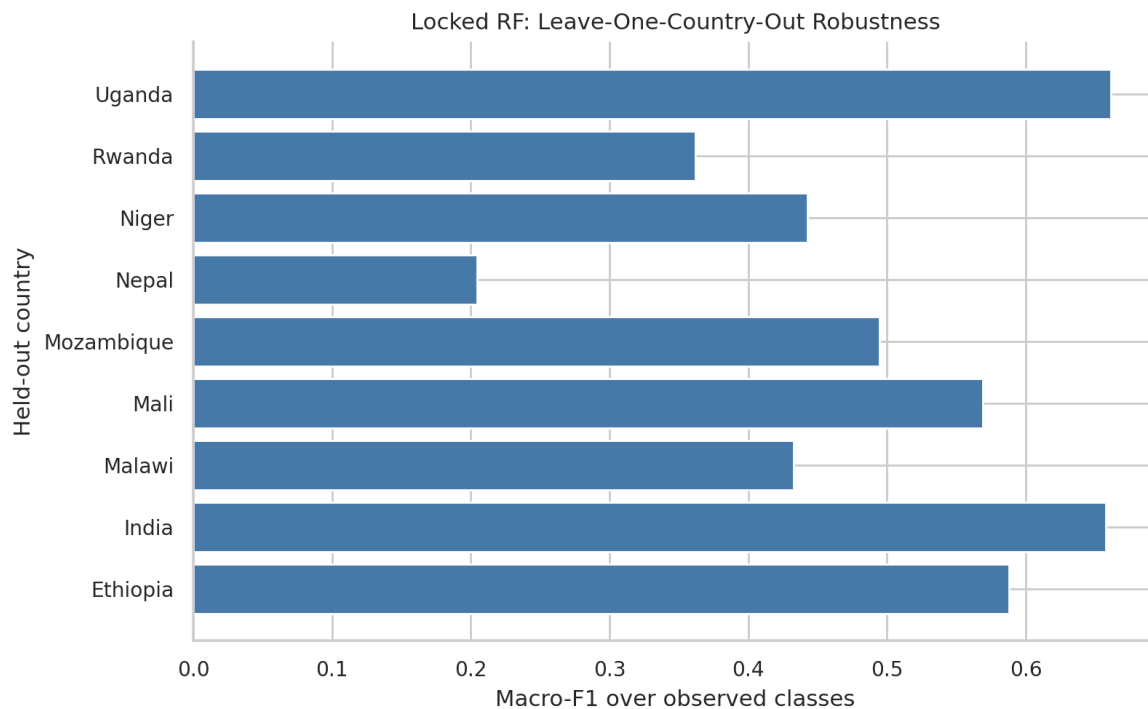


Figure 9.4. Post-hoc country robustness (Source: project export)

Leave-one-country-out results varied widely, with a mean country Macro-F1 around 0.4902 for the locked RF in the saved comparison. This diagnostic asks a harder question than ordinary stratified CV because the held-out country supplies no training examples. It does not supersede the one-time historical holdout metric.

9.9 Post-Hoc Decision

None of the post-hoc alternatives tested provided sufficient evidence or performance improvement to justify replacing the locked Random Forest model.

10. System Design and Implementation

10.1 Overall Architecture

The Next.js browser client sends HTTP/JSON requests to FastAPI. The API validates one raw record and calls the verified serialized pipeline. Dashboard endpoints serve exported JSON evidence.

Figure 10.1 follows the implemented route from a Next.js browser to FastAPI validation and the serialized sklearn pipeline. Dashboard figures follow a separate read-only path from exported JSON evidence through FastAPI endpoints. No database, cloud service or browser-side model execution is involved.

Implemented system architecture

Exported evidence JSON also feeds the FastAPI dashboard endpoints.



Figure 10.1. Current Next.js, FastAPI and serialized-pipeline architecture

10.2 Technology Stack

Python, pandas, scikit-learn, joblib and FastAPI support inference and API delivery. Next.js, TypeScript, React and Recharts support the browser interface.

Table 10.1. Implemented technology stack

Layer	Implementation	Role
Model development	Python / pandas / scikit-learn	Preprocessing, CV and RF
Artifact	joblib + JSON metadata	Locked inference and verification
Backend	FastAPI / Pydantic	Validation, prediction and evidence API
Frontend	Next.js / React / TypeScript	Form, dashboards and charts
Charts	Recharts	Interactive evidence views

10.3 Production Inference Pipeline

The exported artifact contains fixed cleaning, fitted preprocessing, SelectKBest(60) and the locked RF. The frontend does not run model preprocessing.

10.4 Serialization, Metadata and Reproducibility

Metadata records 32 raw predictors, 102 preselection processed columns, 60 selected columns, class order and artifact SHA-256. Export verification reports exact preprocessing and reload parity.

The export manifest records an artifact checksum, 32 raw inputs, 102 preselection columns, 60 selected columns and class order 0/1/2. Table 10.2 identifies the files that make the deployment auditable. Startup checks fail closed if the model file or checksum is invalid.

Table 10.2. Production artifacts and their verification roles

Artifact	Recorded role	Evidence
final_inference_pipeline.joblib	Fitted cleaning, selection and RF	SHA-256 in metadata
final_model_metadata.json	32-input schema and class mapping	102 -> 60 feature counts
export_verification.json	Fresh reload and parity audit	status PASS; tolerance 1e-12
dashboard_assets/	Saved EDA and evaluation evidence	Manifest and CSV/JSON exports

10.5 FastAPI Backend

The backend loads and verifies the trained model once during application startup, ensuring it is ready for prediction requests. For each request, it validates all 32 required input fields before passing them to the model pipeline. The response includes the predicted functionality class, its associated probability, and the probabilities for all three possible functionality classes.

10.6 API Contract and Endpoints

The prediction route accepts explicit nulls in supported fields and safely handles unseen categories. Separate routes expose model metadata and exported dashboard evidence.

10.7 Next.js Frontend

The implemented routes are Dashboard, Predict, Data Insights, Model Insights and About. The prediction page uses a five-step form populated from the API schema.

The current App Router provides five pages with a shared header. Model Insights displays all five Stage-6 baseline families and three finalists separately. The form uses searchable categorical choices, country-scoped admin regions, conditional fields, null transmission and a stale-result notice after edits.

10.8 Prediction Workflow

Users enter known fields, review values, submit one JSON record and see the predicted class and probability distribution. Blank entries are sent as null.

Figure 10.2 traces a request from the five-step form to class probabilities. The screenshot in Figure 10.3 shows the Location stage with country and region selectors; Figure 10.4 shows a completed live smoke test using unknown fields. Its displayed probabilities illustrate the UI contract, not a validation metric.

One prediction request

Blank supported values are transmitted as JSON null.

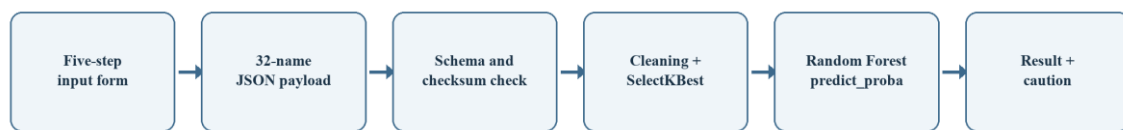
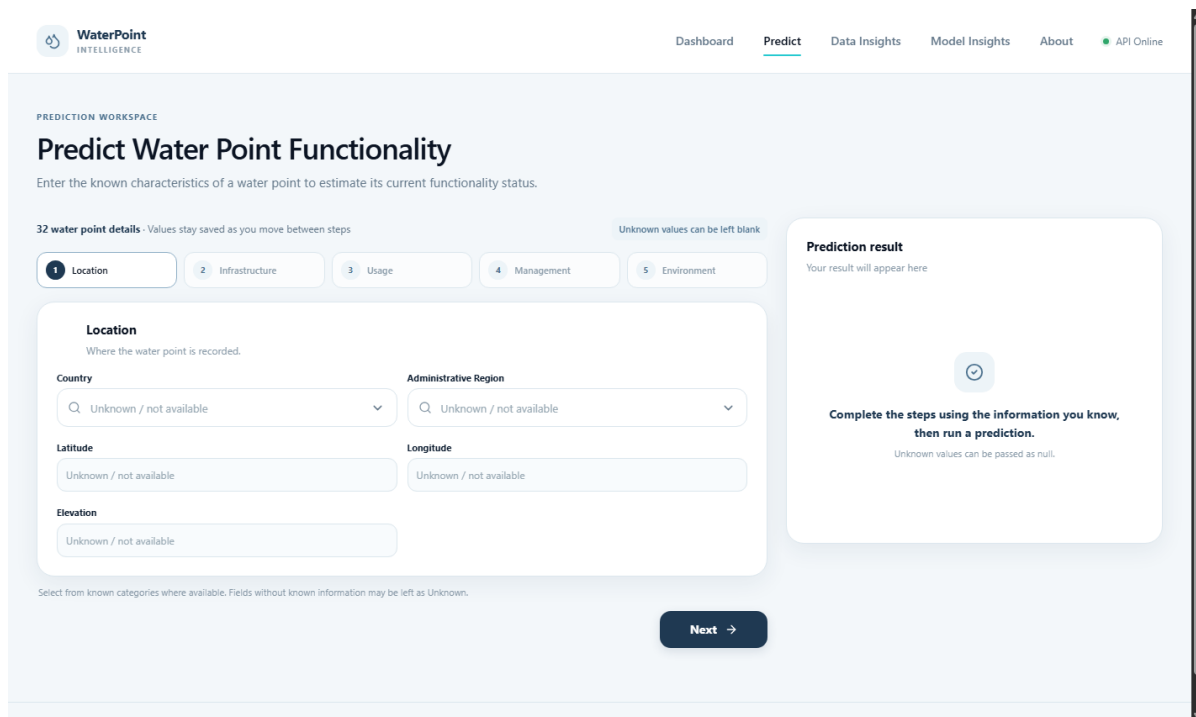


Figure 10.2. 32-field request validation, fitted transformations and result presentation



WaterPoint INTELLIGENCE Dashboard Predict Data Insights Model Insights About API Online

PREDICTION WORKSPACE

Predict Water Point Functionality

Enter the known characteristics of a water point to estimate its current functionality status.

32 water point details · Values stay saved as you move between steps · Unknown values can be left blank

1 Location 2 Infrastructure 3 Usage 4 Management 5 Environment

Location
Where the water point is recorded.

Country
Unknown / not available

Administrative Region
Unknown / not available

Latitude
Unknown / not available

Longitude
Unknown / not available

Elevation
Unknown / not available

Select from known categories where available. Fields without known information may be left as Unknown.

Next →

Prediction result
Your result will appear here

Complete the steps using the information you know, then run a prediction.
Unknown values can be passed as null.

Figure 10.3. Current five-step form collecting the production input contract

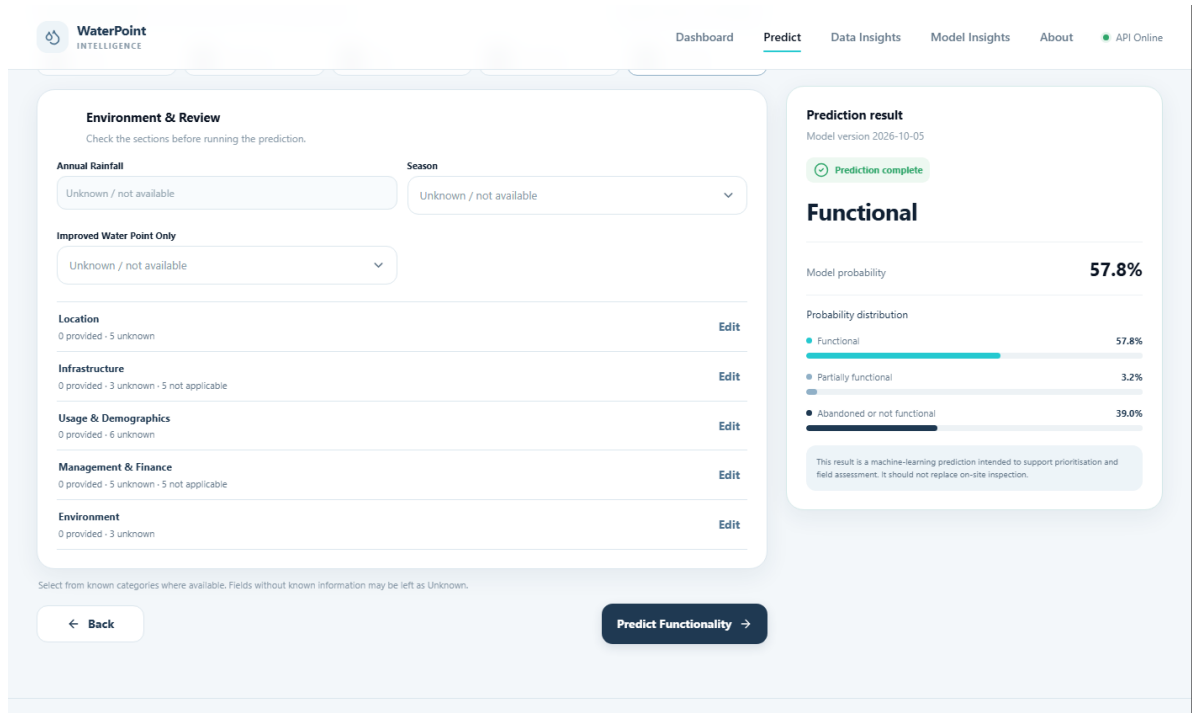


Figure 10.4. Live all-unknown integration check displaying the predicted class and three probabilities

10.9 Dashboard, Data Insights and Model Insights

The frontend presents dataset summaries, class patterns, baseline model families, optimisation finalists and robustness evidence from backend data routes.

Figure 10.5 shows summary cards and country composition loaded from the API. Figure 10.6 shows the final-model panel and five-family baseline comparison. These screenshots were captured from the current production build against the local FastAPI service.

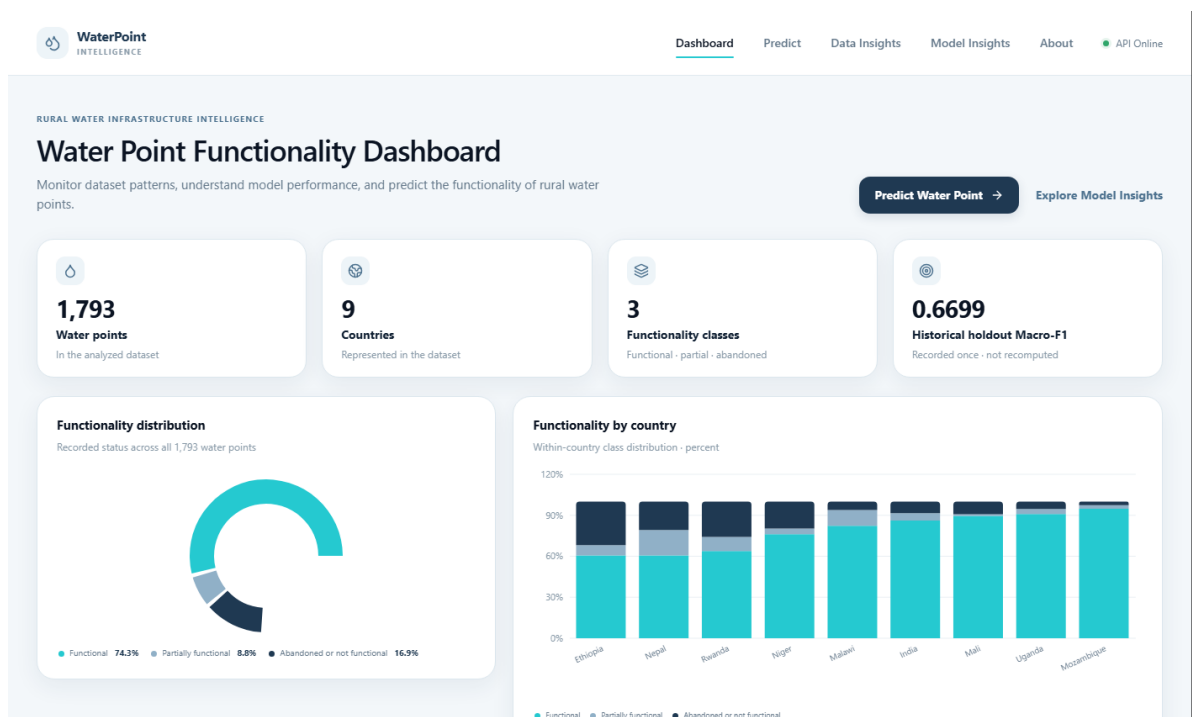


Figure 10.5. Current dashboard with API-fed dataset metrics and country composition

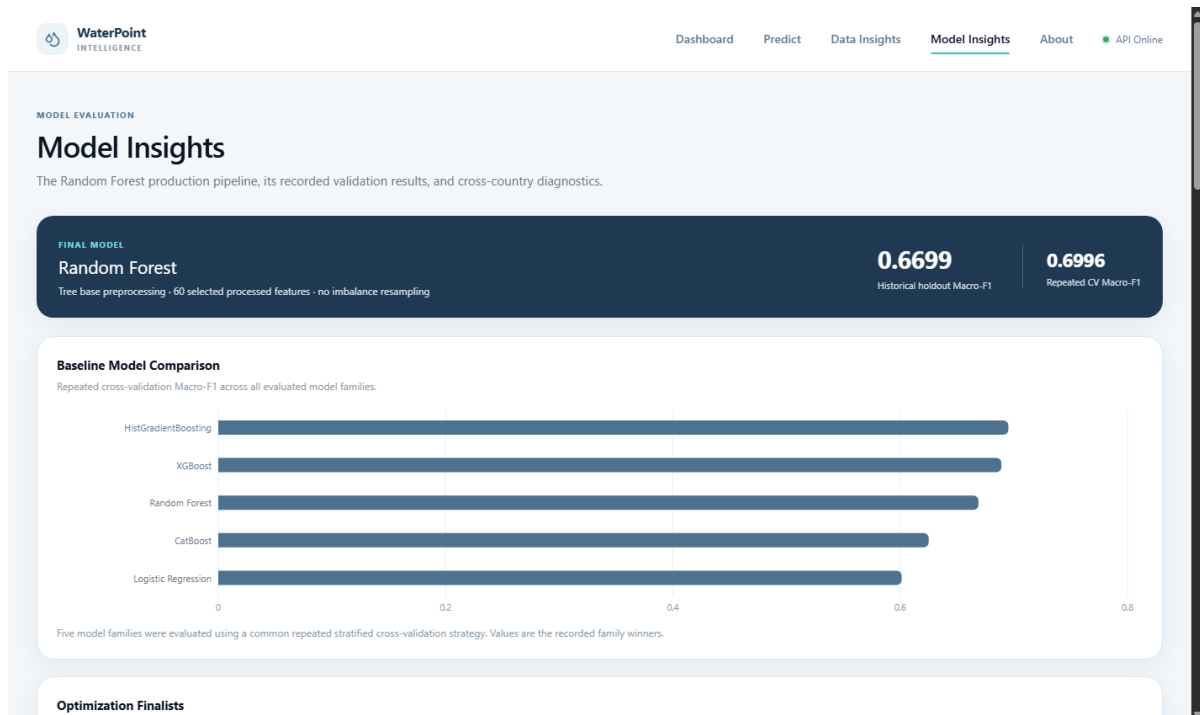


Figure 10.6. Current Model Insights showing the locked RF and five Stage-6 families

10.10 Responsible Prediction Presentation

The result warns that a model probability is not a guarantee and that a field inspection remains necessary.

The result screen presents the predicted label, all three probabilities and a caution that the score does not replace inspection. Presenting the full distribution is useful when the top two classes are close, because a single label hides ambiguity. The values remain outputs of the saved classifier and have not been independently calibrated; the wording should therefore be read as model evidence rather than a probability guarantee about a specific water point.

10.11 Running the System

From the repository root, start `uvicorn backend.main:app --reload`. In `frontend/`, install dependencies and run `npm run dev`. The documented default API base is `http://127.0.0.1:8000`.

Table 10.3. Implemented API routes

Route	Purpose
GET /health	Readiness
POST /api/v1/predict	Predict one 32-field record
GET /api/v1/model-info	Safe model metadata
GET /api/v1/input-schema	Field contract
GET /api/v1/dashboard/summary	Dataset summary
GET /api/v1/dashboard/eda	EDA evidence
GET /api/v1/dashboard/model-performance	Model evidence
GET /api/v1/dashboard/feature-importance	Importance evidence
GET /api/v1/dashboard/robustness	Robustness evidence

11. System Testing

11.1 Testing Objectives

Tests checked schema and class order, serialized-pipeline parity, API validation, null and unseen-category handling, and frontend interactions.

System checks were separated into artifact parity, API behaviour, frontend static quality and live browser flow. Figure 11.1 illustrates the layers. A passing model test does not erase a failing frontend test, so results are reported independently.

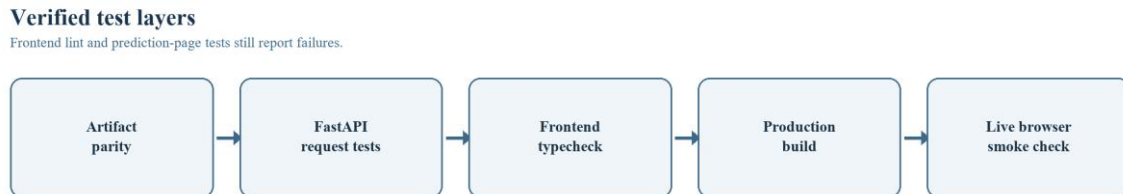


Figure 11.1. Artifact, API, frontend and browser smoke-check layers

11.2 Production Pipeline and Serialization Tests

The saved export verification reports preprocessing parity, 102 preselection features, 60 selected features, fresh-process loading and numerical prediction/probability parity.

Five focused pipeline tests passed. They verify artifact contract, class/probability order, sentinel equivalence, schema rejection and fresh-process reload. The export verification file separately records exact preprocessing parity and probability agreement within absolute tolerance $1e-12$.

Table 11.1. Focused production-pipeline test outcomes

Test area	Verified outcome	Source
Artifact contract	Passed	test_final_inference_pipeline.py
Probability class order	Passed	test_final_inference_pipeline.py
Sentinels and nulls	Passed	test_final_inference_pipeline.py
Schema rejection	Passed	test_final_inference_pipeline.py
Fresh-process parity	Passed	test_final_inference_pipeline.py

11.3 Backend API Tests

The focused FastAPI suite was run against the current source: 17 tests passed, with one Starlette/httpx deprecation warning. Tests cover startup loading, metadata and schema routes, prediction parity, null and unseen-category handling, invalid inputs, dashboard evidence, and failure-closed model verification.

11.4 Frontend Tests and Build Verification

The current frontend passed TypeScript checking and a production build. Its Vitest run reported 9 passing and 4 failing tests, all failures in the prediction-page suite. ESLint reported one error in the prediction page and three warnings. These are open software-quality issues, not evidence that the locked model or historical evaluation changed.

Table 11.2 records the current frontend outcomes. The production build and TypeScript check passed; lint and the prediction-page test suite did not. Those failures are disclosed for correction before submission review, while the report itself leaves the application source unchanged.

Table 11.2. Current frontend checks and open failures

Check	Observed outcome	Interpretation
TypeScript	PASS	tsc --noEmit
Production build	PASS	Five app routes generated
Vitest	9 passed; 4 failed	Failures in prediction-page tests
ESLint	FAIL	One React effect error; three warnings

11.5 End-to-End Integration Tests

The contract connects a schema-driven form to the API and then to the serialized pipeline. No independent live deployment measurement is claimed.

A local browser smoke check loaded dashboard and model evidence, traversed the five-step form, submitted an all-unknown 32-field request and displayed a three-class result. This proves the integration path for that case, not the predictive validity of an all-unknown record. Table 11.3 distinguishes this check from API unit tests and untested deployment conditions.

Table 11.3. Local end-to-end and API validation cases

Case	Observed result	Evidence
All-unknown 32-field UI submission	Class and three probabilities displayed	Live browser smoke check
Valid API prediction	Passed	Backend test suite
Null and unseen category	Passed	Backend test suite
Invalid numeric / missing field / extra field	Rejected as expected	Backend test suite
Savings sentinel 999 and 888	Passed to cleaner as expected	Backend test suite

11.6 Invalid, Missing, Null and Unseen-Category Cases

The request schema preserves all 32 names, allows explicit null values where supported and rejects invalid field values; the model encoder tolerates unseen nominal categories.

Input validation has two different obligations. The API must reject malformed requests, including missing canonical keys, extra fields and invalid numeric values. For a valid 32-field object, the fitted cleaner must preserve the intended meaning of nulls, savings sentinels and previously unseen nominal categories. The focused backend tests exercise both obligations, while the browser smoke check demonstrates only one valid path through the interface.

A valid all-unknown submission is useful for contract testing because it exercises default and null handling end to end. It is not representative of a field survey and cannot support a claim about predictive quality. The result screenshot is therefore labelled as an integration example, and the historical holdout remains the source for model performance.

11.7 Issues Identified and Refinements

Production parity checks addressed training/inference drift. The frontend exposes class distributions and cautions rather than a single unexplained label.

The current frontend checks identify concrete work before a submission-quality release: four prediction-page Vitest cases fail, and ESLint reports one React effect error plus three warnings. TypeScript and the production build pass, and the live form can produce a result, so the failures concern test expectations and static quality rather than an inability to serve the app. They should be resolved in application work and rerun; this report records the observed state without changing source or retroactively treating the checks as passing.

12. Discussion

12.1 What the Results Say About the Problem

Predictors contain meaningful structure, but the small minority class and geographic heterogeneity restrict individual-level reliability.

The gap between 0.8663 holdout accuracy and 0.6699 Macro-F1 illustrates the core task difficulty. Most points are Functional, so a model can perform well on the dominant class while missing many intermediate failures. The data support useful ranking and broad triage patterns, but the held-out class-level evidence does not support unattended decisions for individual assets. The output is best read as a structured second opinion alongside recent inspection records.

12.2 Why Random Forest Was Selected

The RF led repeated development CV, had a smaller train–CV gap than the other finalists and used a reproducible, compact selected representation.

The selected RF improved materially over its own baseline, reducing the train–CV gap by roughly half while increasing Macro-F1. The gain is linked to the tuned tree configuration, rare-category handling and selected processed features as a combined pipeline; the experiment design does not isolate a causal contribution for each component.

12.3 The Partially Functional Class Challenge

Only six of 31 Partial holdout cases were recognised. Most were predicted Functional, making unreviewed operational use particularly risky.

The Partial class is rare and sits between fully Functional and clearly Abandoned conditions. Its 31 holdout examples support only a coarse recall estimate. Since 22 were predicted Functional, operational triage should maintain independent inspection rules for intermittent or borderline service even when the model favours Functional.

Table 12.1. Operational reading of selected error patterns

Observed pattern	Possible consequence	Practical response
22/31 Partial called Functional	Degraded sites may be deprioritised	Use inspection rules alongside score
Country robustness below stratified CV	New-region predictions may transfer poorly	Validate locally before rollout
Uncalibrated probability	Overconfidence in displayed percentages	Present as model scores with caution

12.4 Precision–Recall and Imbalance Trade-offs

SMOTENC increased Partial recall in a later diagnostic while lowering aggregate quality. The choice depends on inspection capacity and the cost of missed degraded sites.

The cost of a false Functional prediction depends on the inspection workflow. Where a missed degraded point delays water access, recall for Partial may deserve greater weight than a single aggregate score. Where field visits are scarce, aggressively raising recall can consume capacity through false alarms. The recorded weighting and resampling runs demonstrate this tension; they do not establish a universally optimal threshold. A future deployment should specify inspection capacity and error costs before choosing any threshold.

12.5 Geographic Information vs Cross-Country Transferability

Location helps ordinary CV, yet leave-one-country-out checks are weaker. Geography may capture context that does not transfer unchanged to new locations.

Coordinates and administrative geography likely capture context not fully represented by technology and management variables. Their usefulness under ordinary CV does not imply portability: country ablations and leave-one-country-out checks show that transfer is weaker. A new region requires local validation before routine use.

12.6 Stakeholder and Operational Value

A probability display can support triage when combined with records and field judgement. It should not automatically allocate repairs or deny inspection.

For a maintenance team, the practical output is a prioritised prompt to investigate, not a replacement for a water-quality or functionality survey. The five form groups make the required context visible, while the three probabilities expose ambiguity between fully functioning, partly functioning and abandoned conditions. A pilot should record whether users changed an inspection decision, what they observed on site and whether the model helped allocate scarce visits more fairly.

12.7 Methodological Reflection

The staged protocol limits leakage and protects the holdout, but a larger external validation set would be needed before operational reliance.

The protocol's principal strength is chronology: baseline comparison, development-only optimisation, final lock, one-time holdout, then post-hoc diagnostics. The principal unresolved methodological need is an independent, recent external set that matches the 32-field production schema.

Table 12.2 summarises the main factors that may affect interpretation of the reported model performance. Internal validity was strengthened by fitting preprocessing transformations within each fold and preserving a separate holdout set, although evaluating multiple candidates during development may still introduce some selection bias. External validity is more limited because country characteristics, data-collection practices, and water-service conditions may vary across locations and over time. Construct validity must also be considered, as the functionality classes are based on recorded survey outcomes rather than a direct and continuous measurement of overall water-service quality.

These threats do not erase the observed improvement over the project baseline. They determine how far the conclusion can travel. The report claims performance for the specified historical evaluation and documents weaker evidence for a new country. Before routine use, a prospective pilot would need a frozen model version, clearly defined outcomes, local class support, review of false Functional cases, and recorded human overrides. Such a pilot would answer an operational question that cross-validation alone cannot answer.

Table 12.2. Validity threats, current controls and remaining checks

Threat	Current control or evidence	Remaining check
Training leakage	Fixed proxy exclusion; fold-fitted transforms	Audit future survey fields
Selection optimism	Common repeated-CV finalist protocol	Fresh external sample
Small Partial support	Per-class counts and confusion matrix	Collect more Partial cases
Geographic shift	Country ablation and leave-one-country-out	Validate each target region
Label and time drift	Historical holdout reported separately	Prospective relabelling and monitoring

13. Ethical, Legal and Operational Considerations

13.1 Decision Support and Human Oversight

Field teams should review predictions and retain authority over inspection and maintenance decisions.

An operational policy should define who may request a score, who reviews it, and how disagreement with a recent field observation is resolved. A high predicted Functional probability must not block a complaint, periodic inspection or urgent response. A local reviewer can combine the model with recent repair records and community reports that were not present in the training data. Recording those overrides would also create useful evidence for later auditing.

13.2 Uncertainty and Automation Bias

The maximum predicted probability is a model output, not calibrated certainty. The UI shows all class scores and a caution.

Model probabilities are uncalibrated scores from the fitted classifier. Displaying all three values can reduce the impression of absolute certainty, but users may still over-trust a prominent class label. Field verification and documentation of contrary evidence remain essential.

13.3 Geographic Fairness and Transferability

Differences in prediction errors across countries may result in missed inspections being unevenly distributed between regions. Therefore, the model should be locally validated before deployment in new countries or regions to ensure its performance remains reliable under different conditions.

The countries contribute uneven numbers of examples and classes. Country-level errors can therefore allocate inspection attention unevenly. This report presents diagnostic differences but makes no formal fairness guarantee; prospective monitoring would be needed in any pilot.

13.4 Data Licensing and Privacy

The dataset source is attributed under the CC-BY terms recorded in the repository. Administrative identifiers and photo fields were excluded from production predictors.

The CC-BY attribution recorded for the source dataset supports reuse with acknowledgement, but the licence does not itself settle privacy or field-safety questions. Precise coordinates and administrative geography can make infrastructure identifiable. A deployed service should limit access to location-bearing records, publish

only aggregated dashboard views where appropriate, and define retention and correction processes for operational data. The prototype does not claim to implement those governance controls.

13.5 Reproducibility and Auditability

Locked metadata, artifact checksum, exported evidence and source notebooks support traceability of reported claims.

The model file, metadata checksum, 32-field schema, selected-feature counts and exported development evidence create a chain from notebook result to API response. Reproducibility still depends on preserving the exact source snapshot, software environment and evaluation chronology. A future revision should receive a new version and independent validation record; it should not overwrite the historical holdout result used in this report.

If the prototype is piloted, an audit record should connect each prediction to the model version, the 32 supplied values, the returned class distribution, the reviewer's decision and the later field observation. Table 13.1 states the minimum record categories. This would allow teams to investigate disagreement and monitor drift while restricting access to site-level information. The current prototype does not claim to maintain such a production audit log.

Table 13.1. Minimum record categories for a future controlled pilot

Record category	Purpose	Handling principle
Model version and timestamp	Reproduce the score	Preserve immutable version ID
32 submitted inputs	Explain the specific request	Limit access to location details
Three scores and label	Review decision basis	Retain all classes
Reviewer action	Document human override	Record reason where practical
Later field observation	Measure real-world error	Separate from training until reviewed

14. Limitations

Table 14.1. Main limitations and implications

Limitation	Operational implication
Only 1,793 records	Some country/class combinations are sparse
Partial recall 0.1935	Many degraded sites may be called Functional
Missing survey fields	Predictions depend on imputation and field availability
Weak cross-country transfer	Unseen-country use requires validation
No established calibration	Probability should not be interpreted as guaranteed risk
No fully independent external set	Historical holdout may not reflect future deployment

The dataset is a historical cross-country snapshot, and the nine countries do not provide equal class support. Stratified folds estimate performance on mixtures resembling the observed data; they cannot prove performance on a new country, a later survey or a different collection process. The leave-one-country-out diagnostic illustrates this limitation but is itself variable because country test sets differ in size and composition.

The Partially functional class has only 31 examples in the historical holdout. A change of a few correctly classified cases would materially change its recall, so the point estimate should not be treated as stable for field

planning. No independent prospective labels, calibrated probability study, operational cost model or controlled user pilot was available. These limits constrain the conclusion to research-stage decision support.

15. Future Work

Future work should collect larger and more recent surveys, define an external country-specific evaluation protocol, assess probability calibration, investigate targeted Partial-class data collection and cost-sensitive triage, and monitor error rates after any controlled pilot. Such work should be treated as a new model-development cycle with a fresh locked evaluation set.

Priority work follows the observed weaknesses: collect more Partial cases and recent surveys; evaluate a fresh external country-specific set; assess calibration and cost-sensitive thresholds; pilot with human review; and monitor errors after deployment. GIS or batch scoring would be useful only after the underlying validation and data-governance questions are addressed.

16. Conclusion

The project delivered an evidence-based multiclass water-point prediction system with leakage-safe evaluation and a verified FastAPI–Next.js inference path. The selected Random Forest achieved 0.6996 ± 0.0481 repeated-CV Macro-F1 and 0.6699 on the one-time historical holdout. Its weak Partial-class recall and geographic transfer limits make it suitable for cautious decision support and further validation, not autonomous field decisions.

The main technical contribution is a reproducible path from audited survey variables to a 32-field inference interface, with development-only selection and an explicitly preserved historical holdout. The result is encouraging for structured prioritisation but limited for the intermediate class and for geographic transfer. Further use should be conditional on fresh local validation and a workflow that keeps field teams responsible for final decisions.

References

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Appendix A. Repository Structure

Primary directories: data/ (raw and processed records), notebooks/ (six executed analysis stages), reports/ (exported results and figures), src/ (preprocessing and modelling), models/ (serialized artifact and metadata), backend/ (FastAPI), frontend/ (Next.js), scripts/ (export and verification), and tests/.

Appendix B. Complete Production Input Schema

Table B.1. The 32 raw predictors required by the production pipeline

No.	Raw predictor
1	country
2	admin1
3	latitude_wp
4	longitude_wp
5	elevation_wp
6	cwfunded_wp
7	wptype
8	pumptype
9	drillmethod
10	piped_source
11	piped_pump
12	rehabyn
13	qtyhh_wp
14	whomanage_wp
15	wp_age
16	rehab_age
17	qtypepeople_wp
18	lockedfullday_wp
19	pumpstrokes
20	wc_present_wp
21	paytocollect_wp
22	improved_wponly_wp
23	qtyhh_c_wp
24	wc_admin_index_wp
25	wc_finance_index_wp
26	wc_mgmt_index_wp
27	wc_maint_index_wp

28	wc_savings_wp
29	wpqty_wp
30	pop_1000
31	annual_rain
32	season

Appendix C. Final Model Configuration

Table C.1. Locked production configuration

Parameter	Value
Preprocessing	tree_base
Rare-category minimum frequency	10
Processed columns before selection	102
SelectKBest mutual information	k=60
Imbalance treatment	none
RF trees	336
Maximum depth	None
Maximum features	0.5
Minimum split size	8
Minimum leaf size	5
Bootstrap	False
Random state	42
Artifact SHA-256	e035c403e4bece3ca1cf7202a10439815a0d5f7396748c0696e1e6258fd984de

Appendix D. Search Configurations and Budgets

The table records the search budget and selected parameters from each saved best candidate. The full best-parameter dictionaries and estimator-level search distributions remain in the optimization evidence and notebook.

Table D.1. Recorded search budgets and selected best-candidate parameters

Family	Draws	Selected best-candidate parameters
HistGradientBoosting	30	learning_rate=0.09677225724712923; max_iter=81; max_depth=None
XGBoost	35	learning_rate=0.08490326871900453; n_estimators=363; max_depth=5
Random Forest	30	n_estimators=336; max_features=0.5; min_samples_leaf=5
CatBoost	12	learning_rate=0.15049772641332726; iterations=413; depth=6

Appendix E. Additional EDA Figures

These supplementary plots preserve numerical and categorical context from Notebook 02 without repeating every exploratory chart in the main discussion.

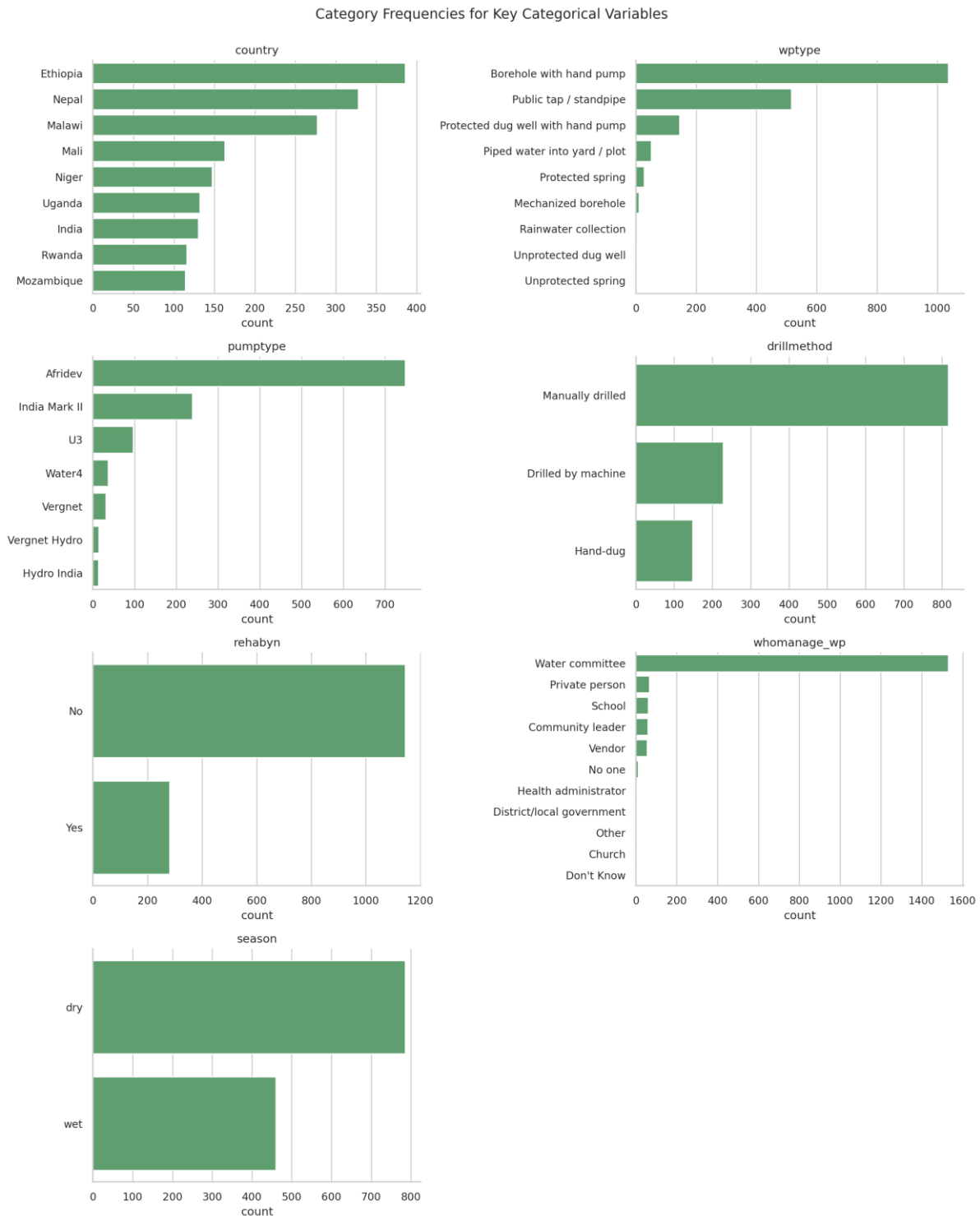


Figure E.1. Frequency of selected categorical levels in the raw survey

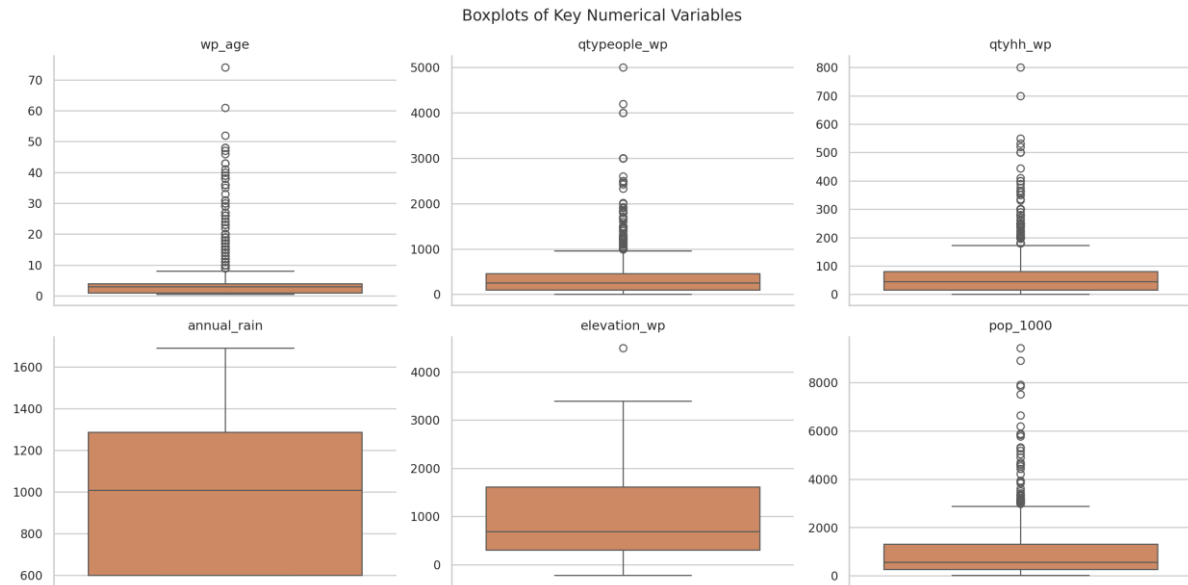


Figure E.2. Numerical boxplots used to review extreme raw values

Appendix F. Additional Model and Robustness Figures

The following exported plots add diagnostic detail to the baseline and post-hoc analyses. They do not change the locked Stage-7 selection.

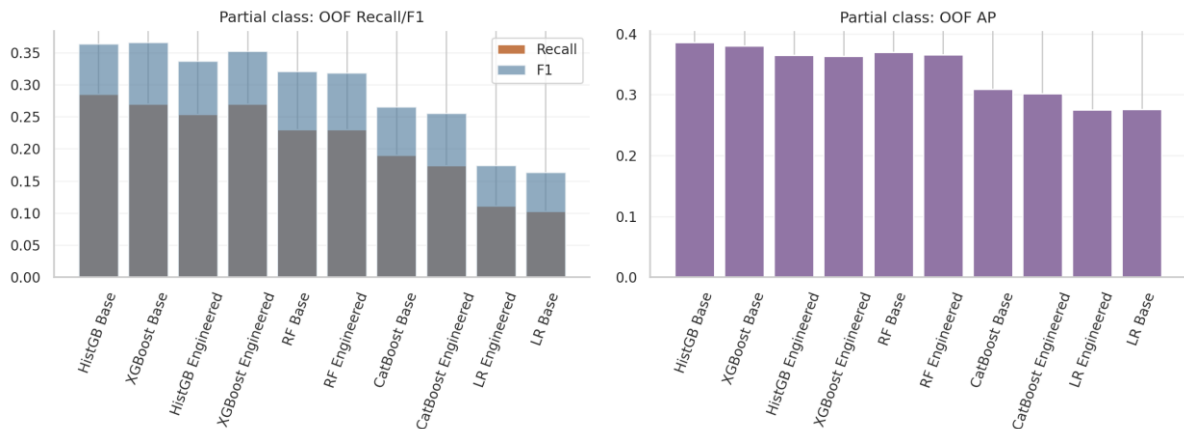


Figure F.1. Stage-6 out-of-fold Partial-class results by model family

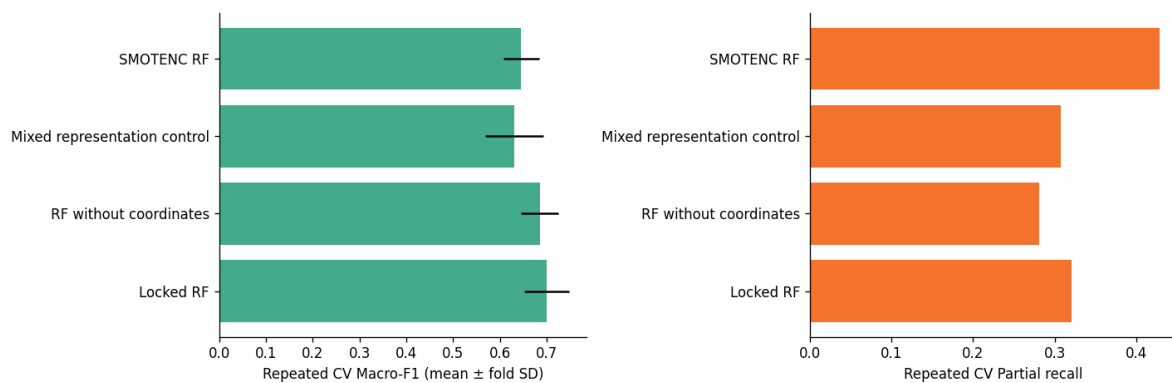


Figure F.2. Post-hoc candidate comparison against the locked Random Forest

Appendix G. API Contract Examples

The prediction endpoint accepts one JSON object containing exactly the 32 canonical field names in Appendix B. Supported unknown values are explicit nulls. The response includes the class identifier, label, maximum model probability, all three class probabilities and model version. The repository example request is `backend/examples/prediction_request.json`.

Table G.1. Prediction API request and response contract

Part	Field or route	Meaning
Request	POST /api/v1/predict	One 32-field JSON record
Request value	null	Known field with unavailable value
Response	prediction.class_id / label	Canonical class mapping
Response	prediction.confidence	Maximum model probability
Response	probabilities	All three class scores
Response	model_version	Artifact version identifier

Appendix H. Testing Evidence

Checks were run against the current repository while preparing this report. The frontend failures are retained as findings rather than silently described as passing.

Table H.1. Executed verification commands and outcomes

Command	Outcome	Scope
pytest tests/test_final_inference_pipeline.py -q	5 passed	Locked artifact
pytest backend/tests -q	17 passed; 1 warning	FastAPI contract
npm run typecheck	Passed	Frontend TypeScript
npm run build	Passed	Next.js production build
npm run test	9 passed; 4 failed	Prediction-page tests fail
npm run lint	1 error; 3 warnings	Predict page effect and lint warnings